

VASTRANGAM

VASTRANGAM GROUP ERP — PLAN OF ACTION

One platform. Every function. Wired together.

The build plan for a multi-company operating system covering a Surat ethnic & western fashion manufacturer running three sister companies — D2C, seven marketplaces, B2B and export on one order book, one stock number and one ledger.

21 modules · 98 apps · 16 apps working today · 82 to build. Built in dependency order, Module 01 first through Module 21 last. Every module is finished — every app on the shared database, verified in a browser — before the next begins.

How to read this. Part I is the integration layer: the law the whole system obeys, the data core that makes it one system rather than 21 programs, and the wiring between modules. Part II is the 21 modules one at a time, each with a diagram of how it actually functions. Every section carries a diagram because the point of this document is that you can see the machine working, not just read a list of features.

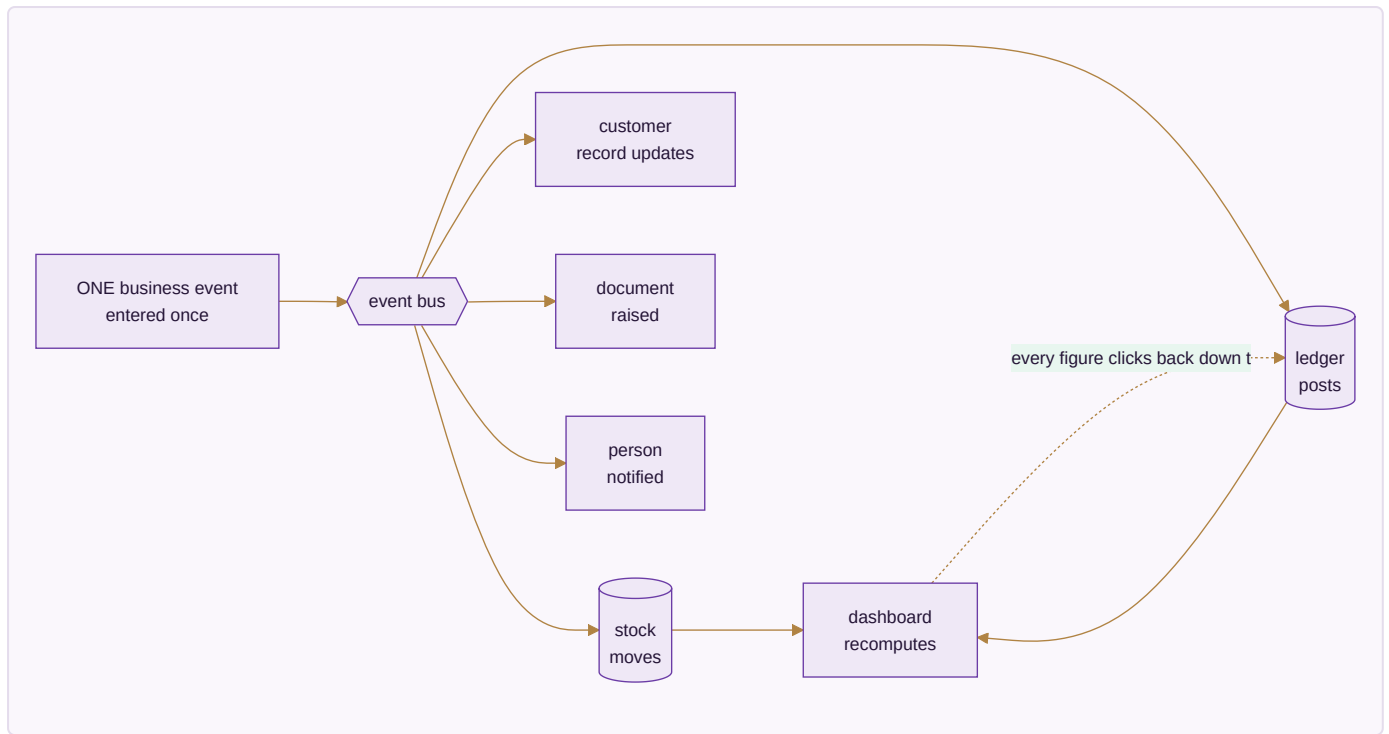
Honest framing. Nothing in this document is described as finished unless it is. Each app is marked **BUILT** (a working file you can open today, carrying its own self-tests) or **SPEC** (designed to this behaviour, not yet written). The count above is the true one: 16 of 98.

PART I — HOW THE SYSTEM WORKS

A0 · THE ONE LAW — INTEGRATION

A business event is entered once and flows everywhere it belongs. No module re-enters data another module already has. No number is maintained separately from the record that created it.

This is the whole design. Everything else in this document serves it.



The test of the law. Any figure on any screen can be clicked down to the record that produced it, and that record down to the document that created it. A dashboard number is a live query over the ledger — never a total someone maintained on the side. If a figure cannot be traced, it is a defect, not a rounding difference.

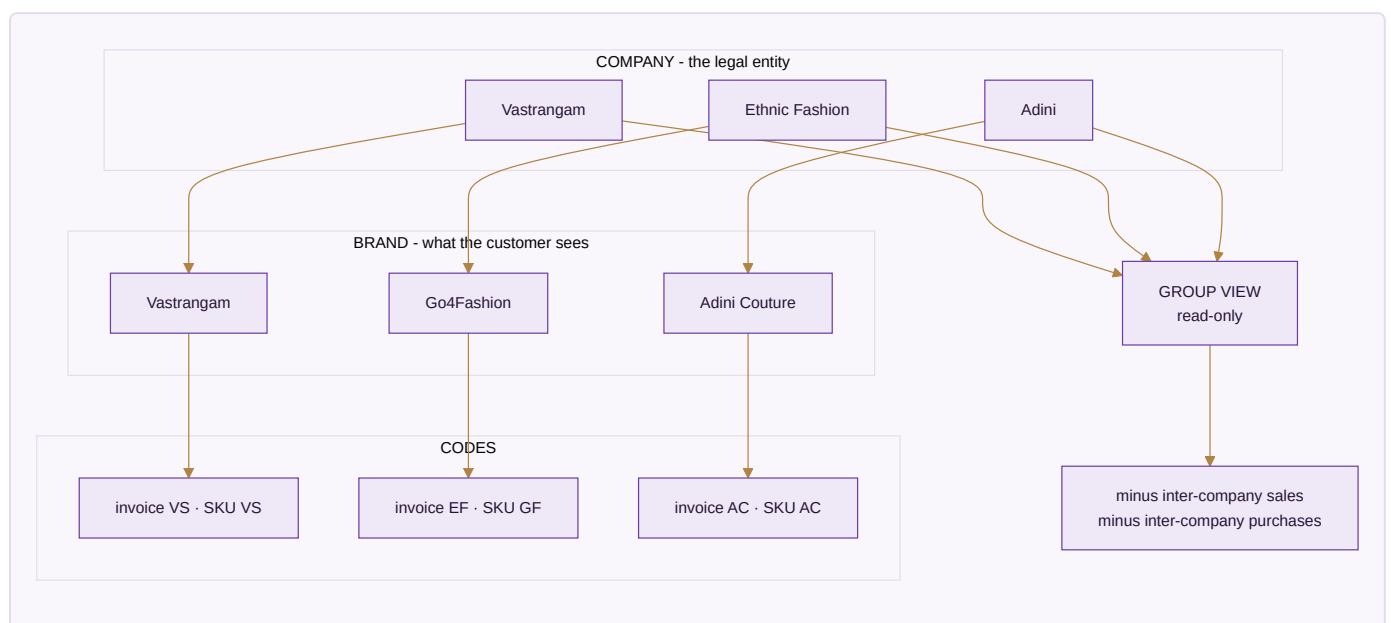
The eight cascades that must fire by themselves

These are the specific chains the platform has to make automatic. If one of them needs a human to re-key something, the system has failed at the only thing that makes it a system.

A single action...	...updates every one of these, in one transaction
A sale , any channel	stock deducted → invoice with GST → ledger (Dr Debtor/Bank, Cr Sales + Output GST) → customer outstanding and lifecycle → settlement expectation → dashboard
A marketplace order pull	sales order created → stock reserved → pick list → fulfilment → on payout, reconciliation and variance check → books
A settlement import	each line matched to its order → commission, fees, TCS, TDS to books → variance → claim raised → bank reconciliation → SKU profit → dashboard
A return	credit note → refund → a wrong return becomes dead stock, never restocked → return cost to P&L → return rate feeds design analytics
A karigar production report	pooled set completion and piece-rate earnings → finished stock in → payout in HR → Karigar Wages posted → cost per piece → design profit
A material purchase	PO → GRN → three-way match → raw stock in → vendor payable → input tax credit → landed cost into COGS
A staff attendance mark	effective-dated salary resolved → payroll → Salaries posted → productivity cost → design cost
A generated image or listing	asset library → product listing on the storefront and each marketplace → marketing calendar → published → campaign return tracked back

A1 · THE THREE COMPANIES

The single most common way a multi-company system goes quietly wrong is collapsing three different ideas into one field. They are kept apart at the root of the database.



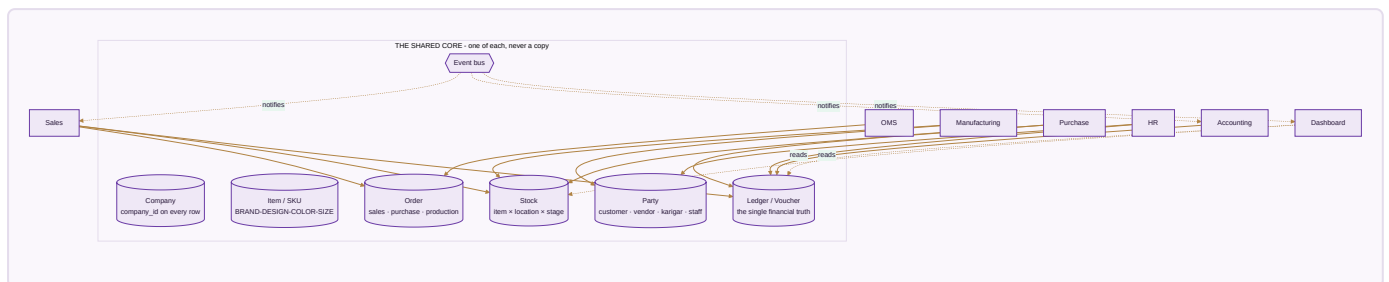
Company ≠ brand ≠ prefix. Ethnic Fashion the company trades as Go4Fashion the brand; its invoices read EF and its SKUs read GF. Reports group by *company*. SKUs use the *brand code*. Invoices use the *company prefix*. Three separate fields, because they genuinely are three different things.

Group is a lens, not an entity. The consolidated view is read-only. Group profit is the sum of the three companies minus inter-company sales and purchases — a real elimination, so moving stock from one pocket to another never inflates group turnover. Editing happens inside a company; the group only reports.

A company without its own registration is still a company. A job-work arm that holds no GSTIN counts in the group figures without being dragged into a return it does not belong in.

A2 · THE UNIFIED DATA CORE

This is the physical reason a sale can touch stock and the books at the same instant: there is one set of master records underneath every module, not one silo per module.



The seven things there is exactly one of. Company · Item/SKU · Party · Stock · Ledger · Order · the event bus that lets modules notify each other. Every business table carries `company_id`, and row-level security enforces it in the database — so even a bug in application code cannot leak one company's data into another's.

Why this matters more than any feature. The predecessor to this system was a set of separate browser tools, each with its own private storage. A sale recorded in the sales tool never reached the accounting tool. Every integration promise in this document rests on that being fixed: one core, one event bus, and the order the sales module writes is the identical row the accounting module reads.

Four rules the core enforces before any module is built

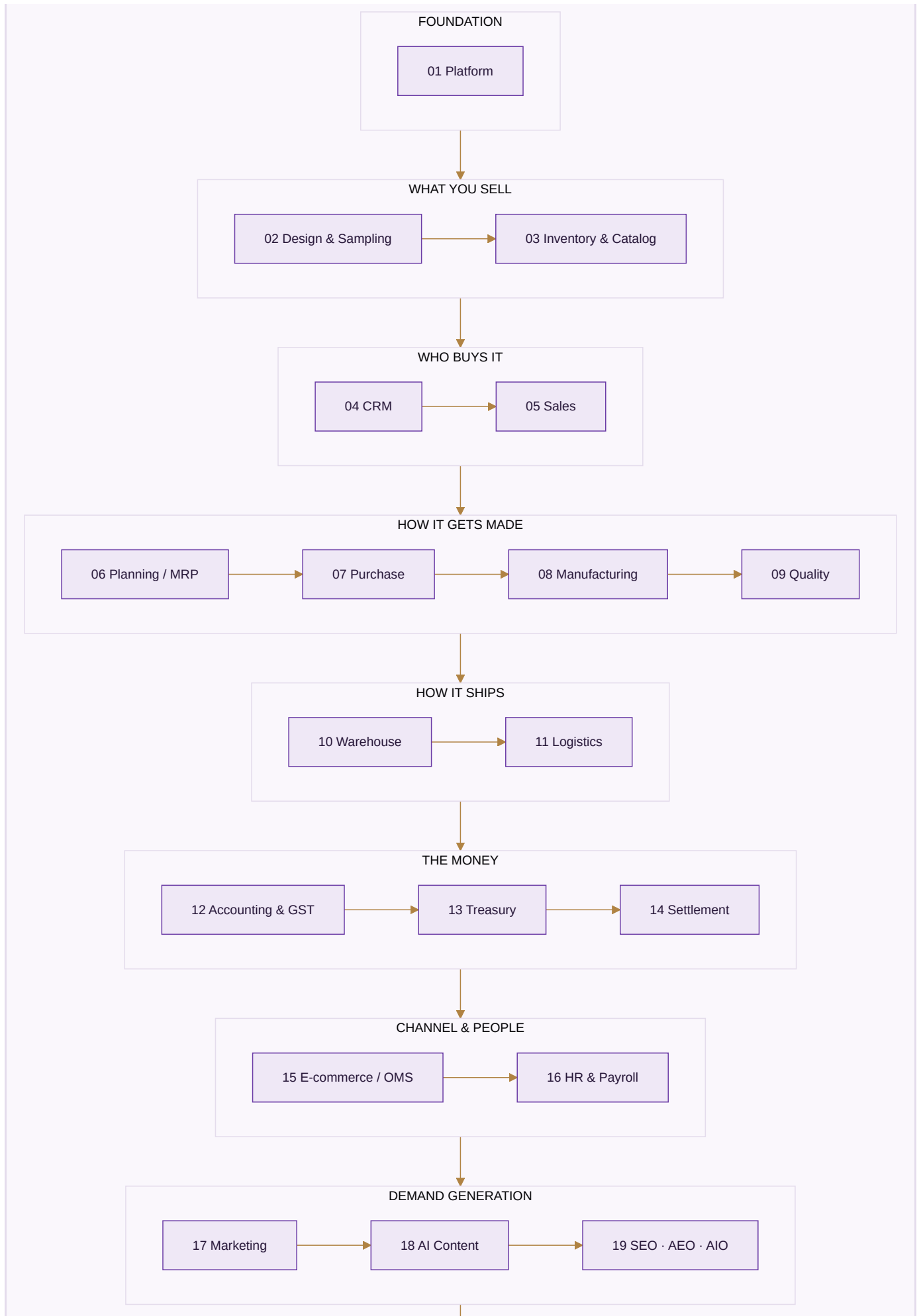
1. **Money is integer paise, never a floating-point number.** Currency arithmetic in floats accumulates error that eventually shows up as a trial balance that will not tie.
2. **Values that change over time are effective-dated** — a salary, a price, a tax rate, a commission. A payroll run for March resolves the salary in force *in March*, not today's. Zero rows in force is an error, never treated as zero.
3. **Nothing is deleted, only deactivated.** A person who leaves is marked inactive; their name is attached to years of earnings, approvals and audit rows that must still resolve.
4. **The audit trail cannot be switched off.** Every edit records who, when, and the value before and after, kept for eight years as the MCA rule requires. It is not a setting that defaults to on — there is no switch, because an

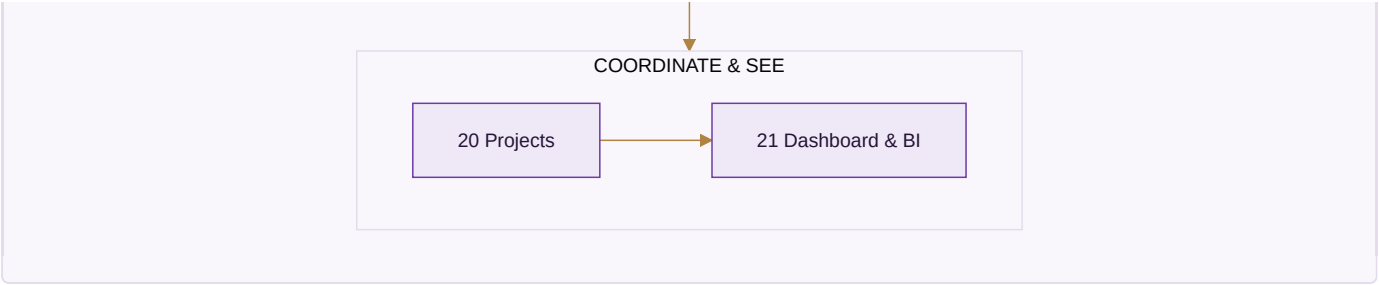
audit trail with an off switch is one that gets silenced exactly when it matters.

A3 · THE MODULE MAP — 21 MODULES

Every module lives inside one application: one login, one company switcher, one data core. The numbering below is the build order, and the build order is dependency order — a module is only built once everything it needs already exists.



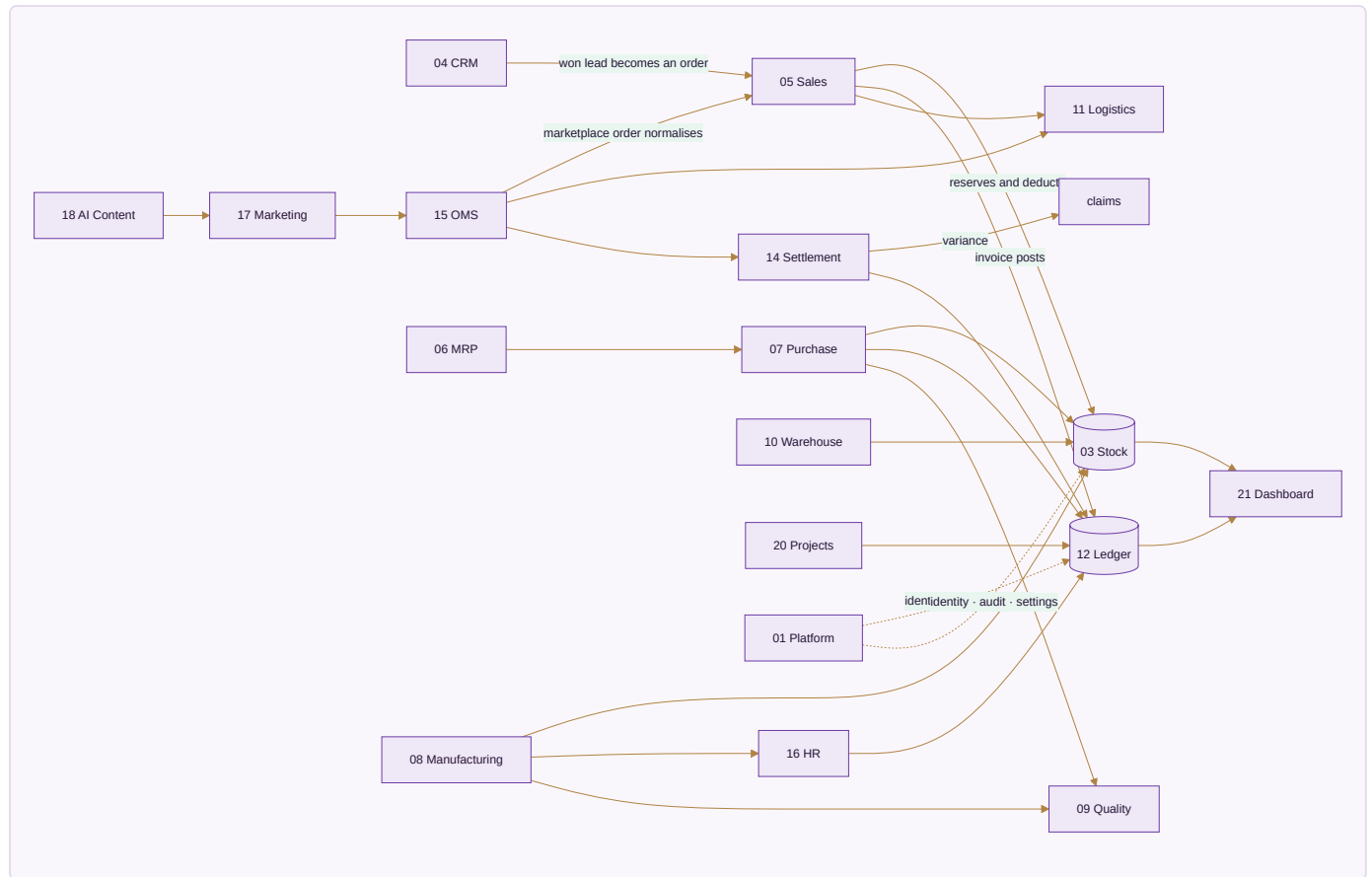




#	Module	Apps	Built	To build
01	Platform	5	1	4
02	Design & Sampling	2	0	2
03	Inventory & Catalog	4	0	4
04	CRM	4	3	1
05	Sales	7	5	2
06	Planning & Requirements (MRP)	3	0	3
07	Purchase	3	2	1
08	Manufacturing	4	0	4
09	Quality & Compliance	2	0	2
10	Warehouse	3	0	3
11	Logistics	5	0	5
12	Accounting & GST	9	0	9
13	Treasury & Financial Planning	3	0	3
14	Settlement	3	0	3
15	E-commerce / OMS	11	2	9
16	HR & Payroll	4	0	4
17	Marketing	7	0	7
18	AI Content Engine	5	0	5
19	SEO, AEO & AIO	3	0	3
20	Projects & Collaboration	6	0	6
21	Dashboard & BI	5	3	2
Total		98	16	82

A4 · MODULE-TO-MODULE WIRING

Every arrow here is a real, required data flow — not an aspiration. This is the fabric that turns 21 modules into one system.



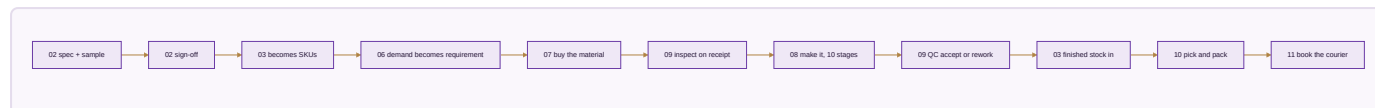
Reading the important arrows.

1. **CRM → Sales.** A won lead becomes a sales order carrying the customer, their price list and their credit tier — nothing re-typed.
2. **Sales → Inventory.** The order reserves and then deducts the single stock number, so the last piece sold anywhere disappears everywhere in the same instant.
3. **Sales → Accounting.** The invoice posts Dr Debtor/Bank, Cr Sales plus output GST, with CGST+SGST or IGST decided from the two GSTINs' state codes — never picked from a dropdown.
4. **OMS → Settlement → Accounting.** Payouts reconcile per line; commission, fees, TCS and TDS post to the books; any variance opens a claim with its evidence.
5. **Purchase → Inventory + Accounting.** The GRN adds stock; the three-way match gates the payable; the GST becomes input credit.
6. **Manufacturing → Inventory + HR + Accounting.** Production adds finished stock, computes karigar piece-rate earnings, posts wages, and rolls cost per piece into design profit.
7. **Accounting → Dashboard.** Every dashboard figure is a query on the ledger, never a separate counter. This is the integrity rule the whole system exists to keep.

A5 · THE FIVE END-TO-END FLOWS

Four of these are the journeys the business actually runs on. Each crosses many modules, which is exactly the point — no single module completes any of them alone.

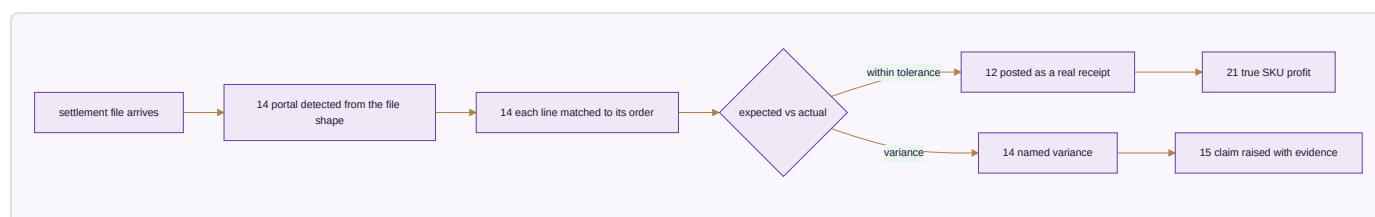
Flow 1 · Design to dispatch



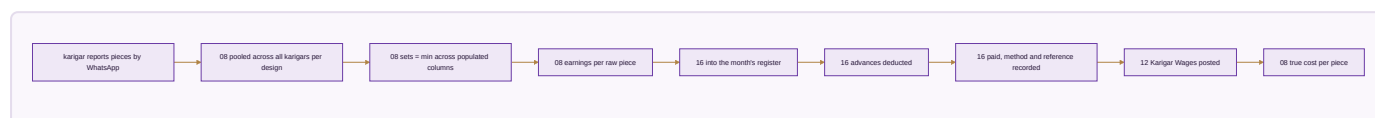
Flow 2 · Order to cash



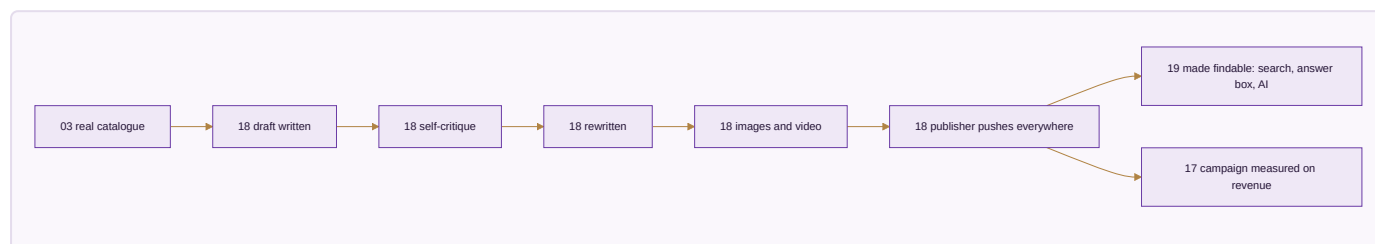
Flow 3 · Settlement to books



Flow 4 · Karigar to payroll



Flow 5 · Content to published



A6 · WHY THIS BUILD ORDER

The order is not a preference. Each module needs something the previous one produces, and building out of order means building against data that does not exist yet.

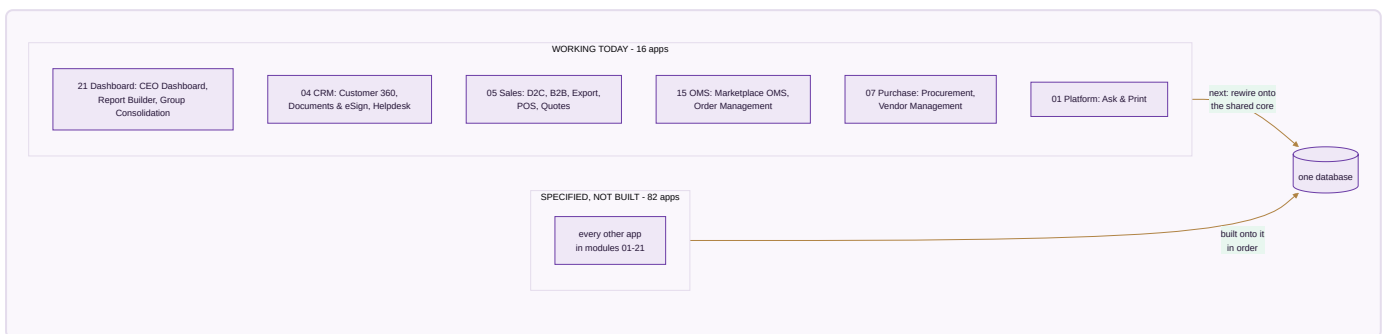


Four decisions worth explaining, because they differ from the obvious arrangement.

- **Platform is first, not last.** Identity, permissions and the audit trail are what every other module writes through. Retrofitting an audit trail means going back through every module.
- **Design & Sampling comes before Inventory.** A product must be specified and approved before it can exist as a catalogue record. Building the catalogue first means inventing styles with no real specification behind them.
- **Quality is its own module, not a step inside Manufacturing.** Rejecting goods at receipt and rejecting goods on the floor are the same discipline, and the certificates that prove a standard is followed belong beside the inspections that back them.
- **Dashboard is last.** It has nothing true to show until the other twenty produce real records. A dashboard built first can only display invented figures.

What a module does when a later module is not built yet. It is finished against the data that exists on its day, and shows more as later modules come online. Module 21's dashboard is real from the moment it is built and grows richer as Module 12's ledger fills.

A7 · THE HONEST STATE TODAY



What "built" means here. Sixteen single-file apps that open in a browser, carry their own self-tests, and pass a full click-through audit with zero console errors in both editions. That is verified, not claimed.

What is honestly not done. Those sixteen apps still run on their own storage. The first work of each module is rewiring its built apps onto the shared core so they read and write the same records as everything else. Until that happens they are good tools, not yet one system.

PART II — THE 21 MODULES

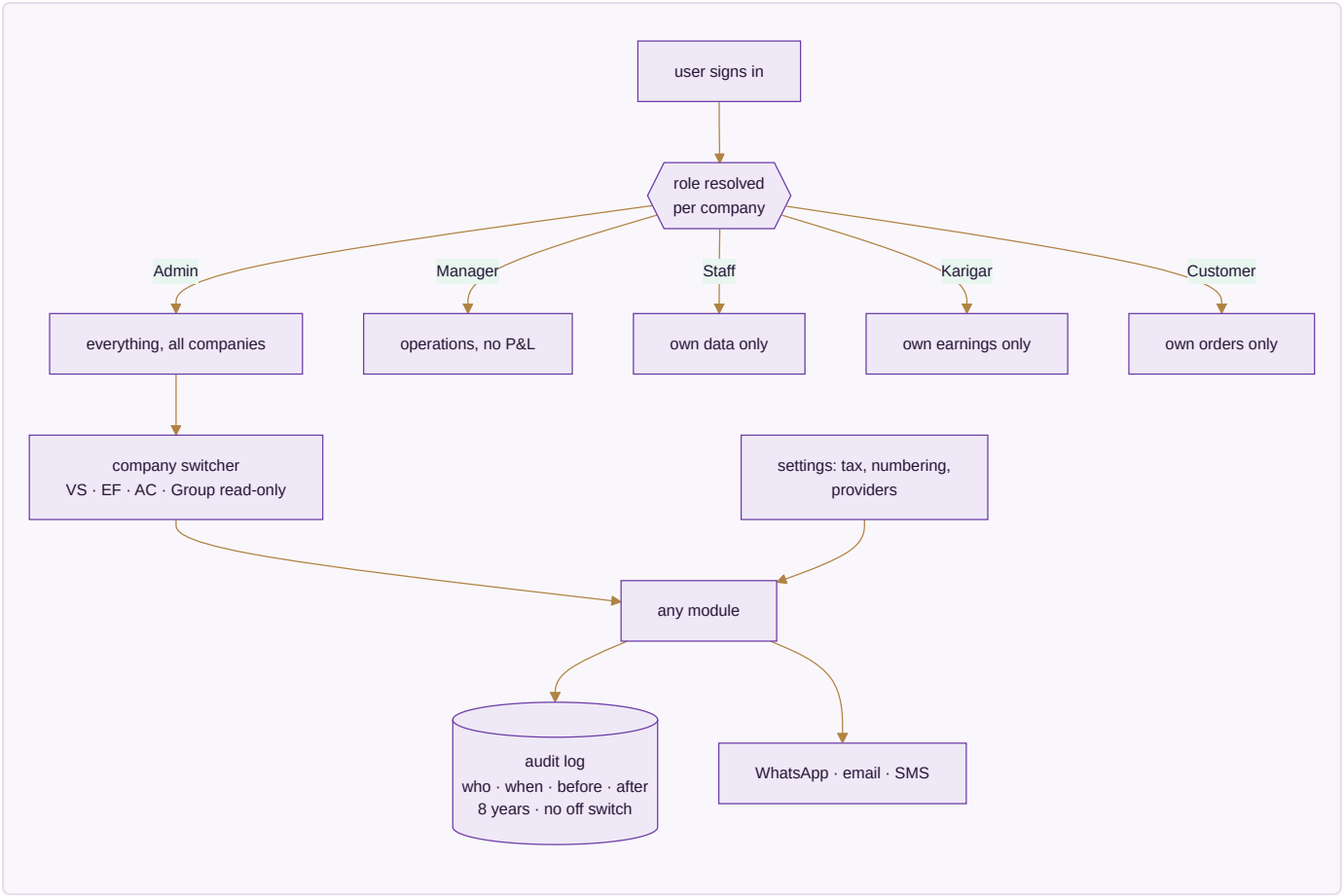
Each module below carries: what it is, a diagram of how it functions, its apps with an honest built or spec mark, the data it owns, the rules that actually govern it, what it reads and writes, and the condition that decides it is finished.

MODULE 01 · PLATFORM

The spine every module runs on

What it is. Not a module you open — the layer underneath the other twenty. Who can see what, how the business is configured, and an unalterable record of everything that ever happened. It is built first because every module above it writes through it, and an audit trail added later is an audit trail with a hole in it.

How it works



The apps (5)

App	State	What it does
Identity, Settings & Audit	SPEC	Users, per-company per-role permissions, company switcher, tax and numbering setup, provider config with an integration-health view, and the browser over the audit trail
Ask & Print	BUILT	Ask from a phone — a ledger, a bill, today's packing slips — and get a PDF back, or print at the office with nothing plugged into the phone
Communications	SPEC	WhatsApp command console, broadcasts, email and SMS, and the scheduled jobs that carry a nudge without anyone remembering to send it
Data Privacy & Consent	SPEC	Consent captured where it is given and honoured downstream; retention and erasure tracked as the two different policies they are
Payment Data Scope	SPEC	The written statement of which systems ever see a card or bank credential — and which never do

What it owns. `companies` · `users` · `user_companies` · `audit_log` · `integration_errors` · `settings_environment` · `whatsapp_messages` · `whatsapp_broadcasts` · `email_campaigns` · `notifications` · `consents`

The rules that matter

- Permissions are a matrix — per company, per role — not one global level. A group of sister companies is exactly where a single global level leaks data across a boundary it should not cross.
- The audit trail has no off switch. Eight years, before-and-after values, MCA rule.
- Every capability runs through at least three interchangeable providers. **A provider named as the source of a figure is a bug** — providers move messages and money; the ledger originates numbers.
- Broadcasts go through the official API, warm up at 200/day, and honour a STOP keyword.
- **This system never asks anyone for a marketplace, bank or account password.** A tool that asks for those has become the thing it should protect its users from.

Reads ← every module · **Writes** → every module

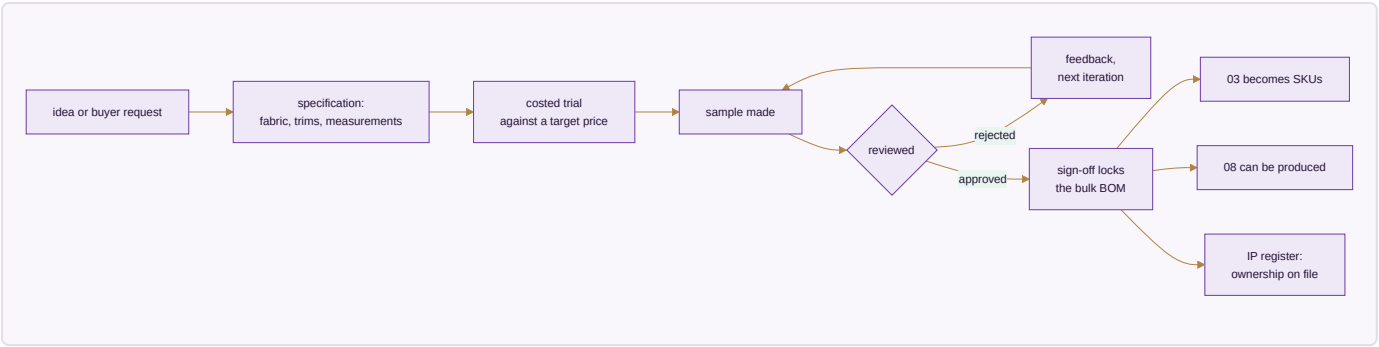
Done when. A karigar sees only their own earnings, an admin switches all three companies and every figure changes with the company, and every edit made during the test is already in the audit browser.

MODULE 02 · DESIGN & SAMPLING

A style exists on paper before it exists as stock

What it is. Where a product moves from a first idea to something the business can actually make and sell — specification, sample rounds, costed trials, sign-off — before it is ever entered as a catalogue record. Building this before Inventory means the catalogue is never populated with styles that have no real specification behind them.

How it works



The apps (2)

App	State	What it does
PLM & Development	SPEC	Specification → sample rounds → costed trials → sign-off, every version kept, so last season's costing is still there
Design / IP Register	SPEC	Trademark and copyright status, the date a design was first shown, and a flag when a near-identical listing appears elsewhere

What it owns. designs (the development record) · samples · sample_iterations · design_ip_register · tech_packs

The rules that matter

- Every version is kept, never overwritten. A design's cost history is the evidence behind this season's price.
- Sign-off is the event that locks the bulk bill of materials. Sample BOMs and bulk BOMs carry different wastage percentages and are not the same record.
- A design with no ownership record on file is a design nobody can defend — which matters most for exactly the designs Module 18 is built to promote at scale.

Reads ← CRM · Writes → Inventory & Catalog · Manufacturing

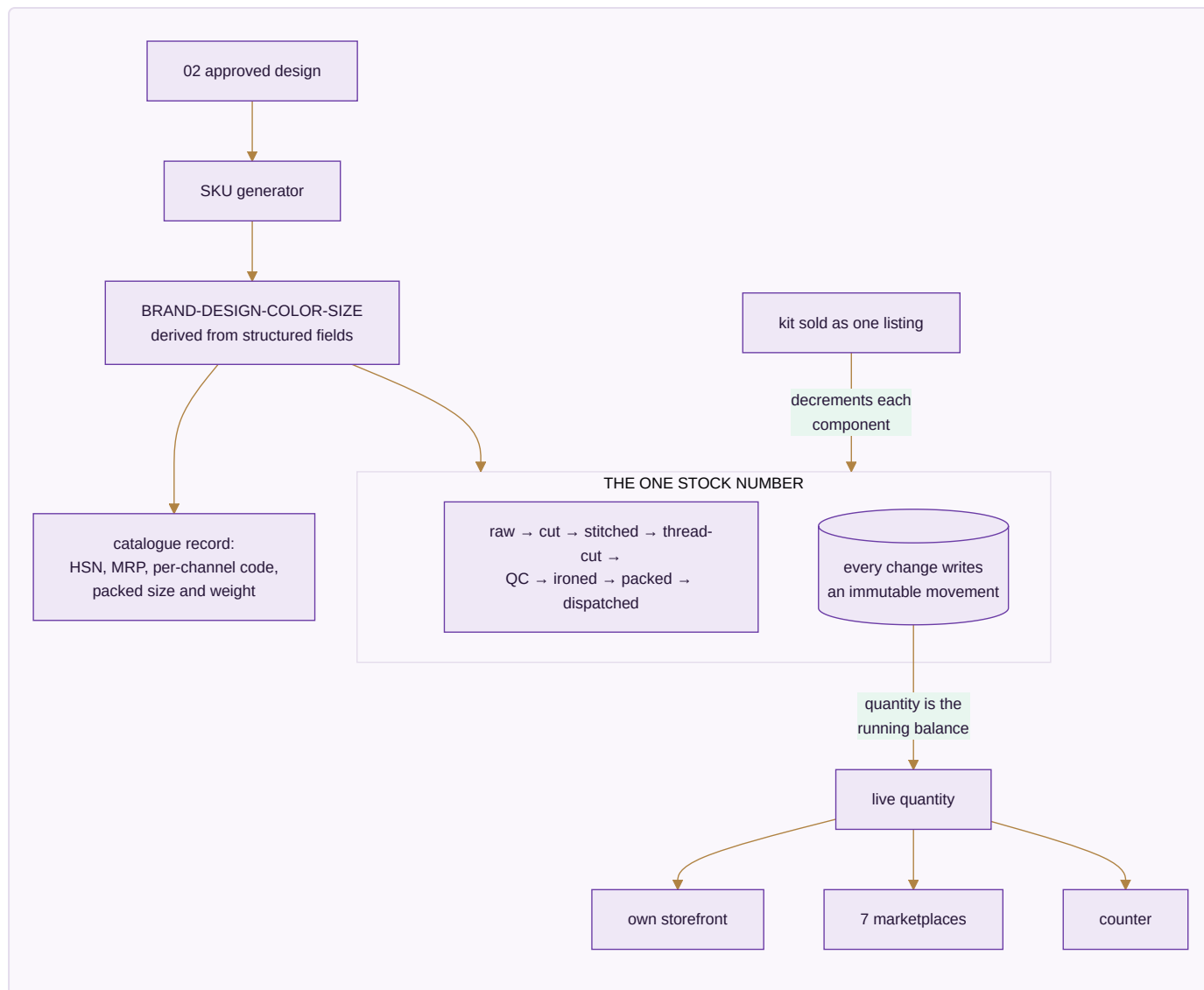
Done when. A design goes idea → sample → rejection → second sample → approval, and only the approved version can generate SKUs.

MODULE 03 · INVENTORY & CATALOG

One number everyone trusts

What it is. The most important number in the system: one quantity per SKU, per location, per stage — read and written by every other module. And one product record that every channel lists from.

How it works



The apps (4)

App	State	What it does
Stock	SPEC	Live quantity by SKU, location and stage, with reorder alerts, batches and dead-stock ageing
Catalog / PIM	SPEC	One product record with each channel's own code mapped to yours, and the packed size and weight that decide courier rate and settle weight disputes
Kit & Combo SKU	SPEC	A sellable SKU made of component SKUs; selling it decrements every component
Master-Data Hygiene	SPEC	Duplicate detection and merge across customers, vendors and designs

What it owns. designs · colors · sizes · items · item_aliases · kit_items · stock · stock_movements · batches · opening_stock · hsn_codes · gst_rates · locations

The rules that matter

- **Movements are the ledger; the quantity is its running balance.** Every transition writes an immutable row, so "how did we get to 4?" is always answerable.
- Stock is event-driven and **not per channel**. The last piece sold on one marketplace leaves every other in the same instant — not three hours later as a cancellation, because cancellations are what a seller rating is lost to.
- The SKU string is *derived* from structured fields. Search, grouping and analytics use the fields, never substring-matching on the string.
- Negative stock is a data fault, not a state the business can be in. An issue beyond what exists is refused and recounted.
- A wrong return is dead stock and is **never added back**.
- Stock valuation sets the balance-sheet figure — the two must be equal, or one of them is wrong.

Reads ← Design & Sampling, every module · **Writes** → every module

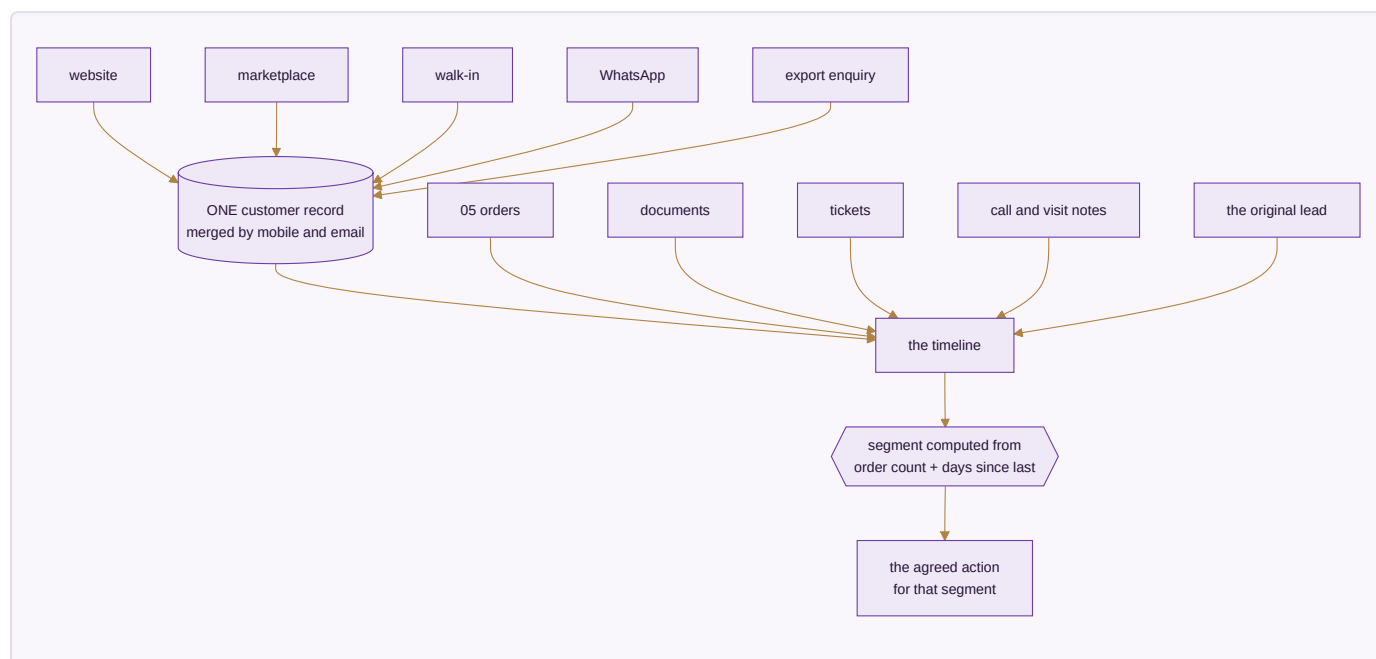
Done when. Stock is one number across every channel, a kit sale decrements all components, and stock valuation equals the balance sheet.

MODULE 04 · CRM

Know every customer completely — and answer them fast

What it is. One record per customer carrying every lead, order, return, document and conversation, whichever channel it arrived on. Whoever picks up the next question can already see everything that came before it.

How it works



The apps (4)

App	State	What it does
CRM & Customer 360	BUILT	Lead to won, then the full lifetime — orders, returns, value, and what to offer next, on one screen
Documents & eSign	BUILT	Every agreement and certificate filed against the record it belongs to; send for signature and the signed copy files itself back
Helpdesk & Live Chat	BUILT	Questions become tickets tied to the order they are about, with the whole history already open
Forms & Feedback (NPS)	SPEC	Post-delivery feedback attached to the <i>design</i> , so a complaint-prone design surfaces as a pattern

What it owns. `customers` · `customer_addresses` · `customer_interactions` · `customer_lifecycle_events` · `loyalty_ledger` · `documents` · `tickets` · `nps_responses`

The rules that matter

- Nobody tags anybody. A customer's segment is computed from two facts only — how many orders and how long since the last — so a buyer moves segment by themselves the moment they buy or go quiet. Hand-tagged segments are always six months out of date.
- D2C, B2B, export, walk-in and WhatsApp merge to one customer by mobile and email. Marketplace customers stay separate because the channel does not share their details, but tie by pattern.
- The pipeline advances Lead → Qualified → Quoted → Negotiation and stops there; won or lost is a separate deliberate act.
- Feedback attaches to the design, not only the buyer — that is what turns a scatter of complaints into a fault Manufacturing can fix.

Reads ← every module · **Writes** → Sales · E-commerce/OMS · Marketing

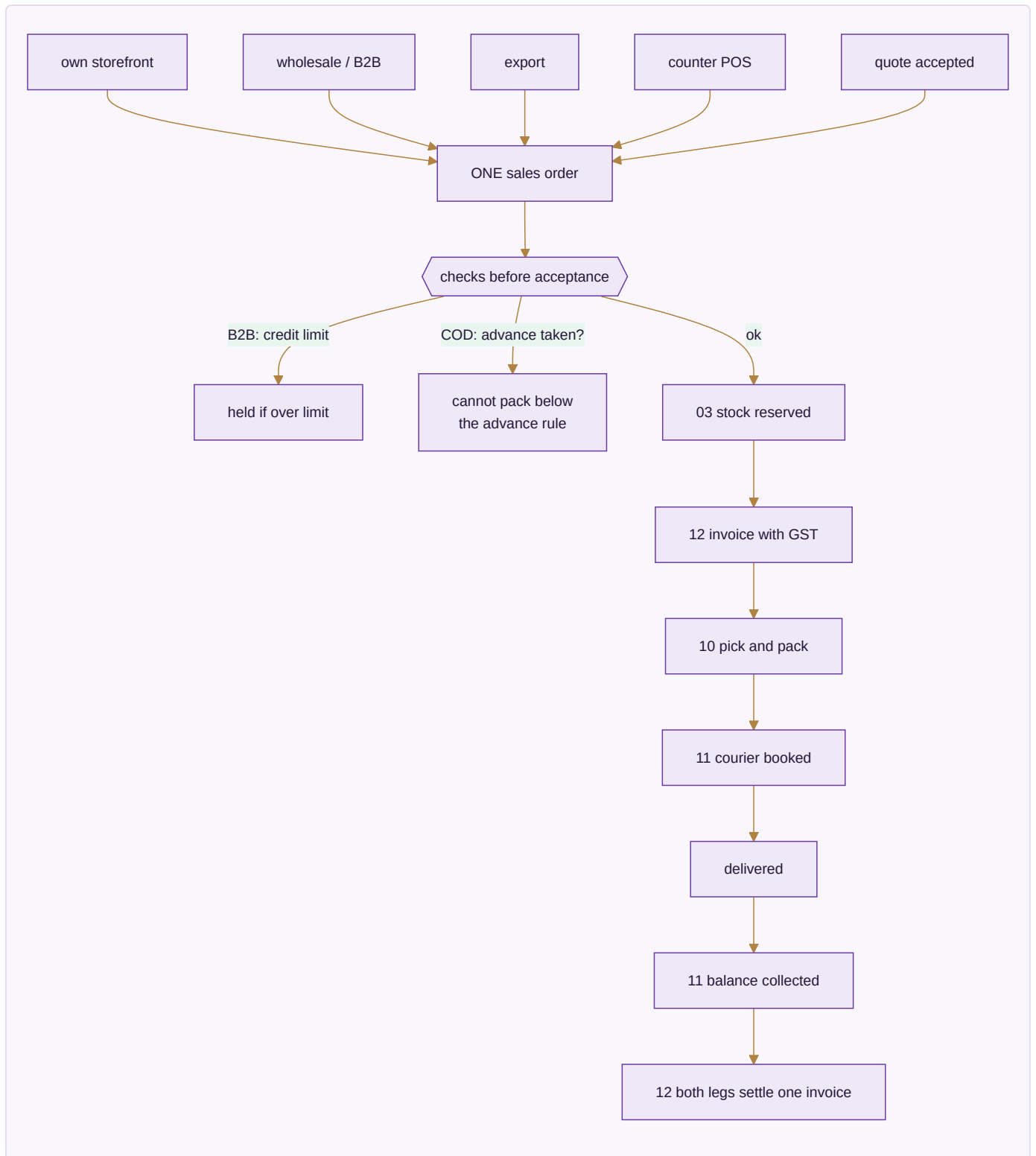
Done when. One customer's whole cross-channel history is on one screen, and crossing an order threshold moves their segment without anyone touching it.

MODULE 05 · SALES

Every way you sell, one order book — to the doorstep

What it is. Counter, wholesale, export and your own website all writing to the same order and drawing on the same stock number. A sale is not finished when it is billed — it is finished when it is delivered and the cash-on-delivery money is in.

How it works



The apps (7)

App	State	What it does
D2C Sales	BUILT	Own-storefront orders cart to dispatch, with loyalty and partial COD
B2B & Credit	BUILT	Wholesale orders with credit limits, tier pricing and outstanding ageing
Export	BUILT	Commercial invoice, packing list, LUT bond and IGST-refund tracking
POS	BUILT	Counter billing drawing on the same stock as the website
Quotes & Proforma	BUILT	Send a quote, convert it to a confirmed order in one step
Couriers & AWB	SPEC	Compare couriers on the order, print the label, follow the AWB to the door
Subscriptions	SPEC	A schedule that raises its own invoice and follows up when a payment fails

What it owns. `sales_orders` · `sales_order_items` · `invoices` · `invoice_items` · `b2b_orders` · `b2b_credit_ledger` · `export_orders` · `customization_orders` · `subscriptions`

The rules that matter

- **A COD order cannot be packed until it carries its advance.** A refused COD parcel costs the courier fee both ways and comes back handled; a customer who has paid something almost always accepts. This one rule is the cheapest defence the business has.
- **Partial COD reconciles both legs to one invoice** — the advance taken online and the balance the courier remits.
- Credit is checked *before* acceptance, not discovered later as a bad debt. Reminders at -3 days, +1 day, and a +7-day soft block.
- The delivery date is **derived** — cut-off plus transit from the warehouse that actually holds stock — never typed. An order no warehouse can serve gets no date at all, because a promise nobody can keep is worse than no promise.
- Export lines are zero-rated under LUT; FX difference between billing rate and realisation rate posts to FX gain/loss.

Reads ← Inventory & Catalog · CRM · Warehouse · Logistics · **Writes** → Inventory & Catalog · Accounting & GST · Warehouse · Logistics

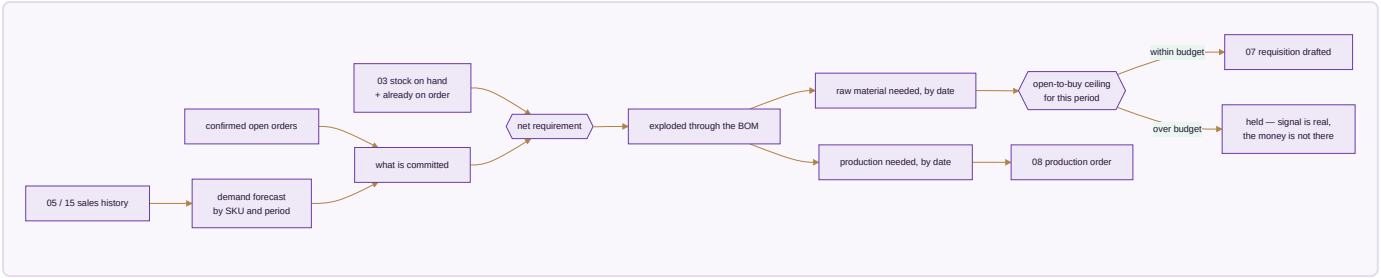
Done when. A storefront order appears within a minute with stock reserved and an invoice raised, and a partial-COD order reconciles both legs by itself.

MODULE 06 · PLANNING & REQUIREMENTS (MRP)

Turn what is selling into what to buy and make

What it is. The module that sits between demand and supply. Confirmed orders and sales history become a requirement before Purchase buys anything or Manufacturing starts anything — otherwise both are guessing.

How it works



The apps (3)

App	State	What it does
Demand Forecast & Signal	SPEC	What sold, by SKU and period, turned into a short-term forecast — the number every downstream requisition is measured against
Requirement Explosion (MRP run)	SPEC	Demand exploded through the bill of materials into what to buy and what to make, and by when
Open-to-Buy / Budget Ceiling	SPEC	A spending ceiling per period that a demand signal alone cannot exceed

What it owns. demand_forecast · mrp_runs · mrp_requirements · open_to_buy_budgets

The rules that matter

- Net requirement is demand minus stock on hand minus what is already on order. Buying against gross demand is how a business ends up holding two seasons of the same fabric.
- Every requisition traces back to the run that produced it, and that run back to the demand behind it. A purchase order whose only justification is "stock looked low" is exactly what this module exists to eliminate.
- **The budget ceiling is a hard guardrail, not a warning.** A genuine demand signal can still commit more money than the business decided to risk this season, and without a ceiling it will.

Reads ← Sales · E-commerce/OMS · Inventory & Catalog · **Writes** → Purchase · Manufacturing

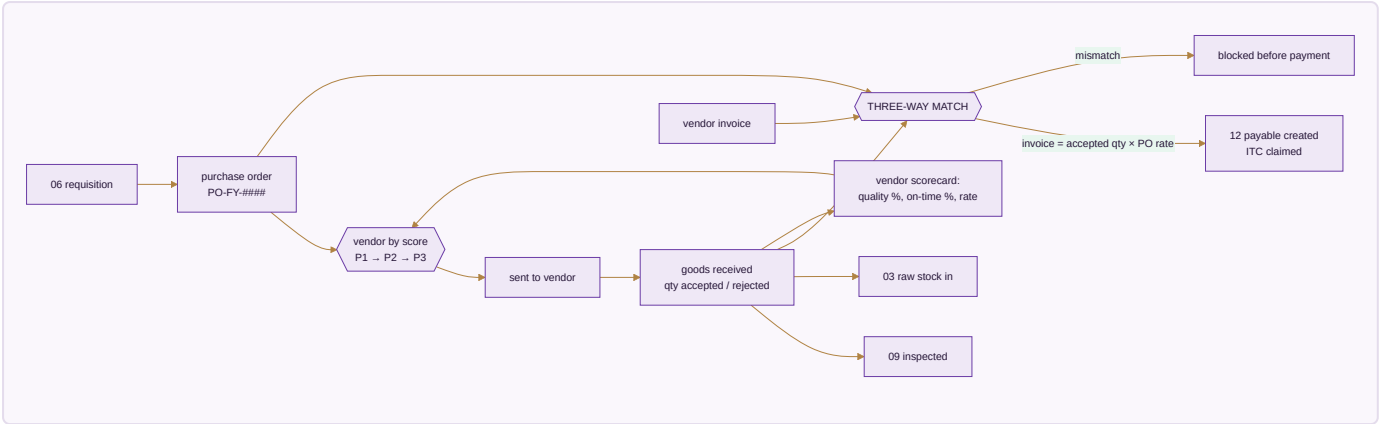
Done when. An MRP run turns a month of real demand into requisitions and production orders that each trace back to the orders that caused them, and the ceiling blocks the run that exceeds it.

MODULE 07 · PURCHASE

Nothing over-billed gets paid

What it is. The buy side end to end — mills, dyers, job workers, packing suppliers — and the control that stops you paying for goods you rejected.

How it works



The apps (3)

App	State	What it does
Procurement	BUILT	Requisition → PO → GRN, with a strict three-way match before any bill is paid
Vendor Management	BUILT	Vendor 360, payables, ageing, a real risk score, and sourcing that follows performance
Insurance Register	SPEC	Cover over stock in transit, stock in the warehouse and product liability, matched against what is actually moving

What it owns. purchase_requisitions · purchase_orders · purchase_order_items · grn · grn_items · vendor_invoices · three_way_match · vendors · vendor_materials · third_party_services · insurance_policies

The rules that matter

- **The three-way match is the whole point.** You ordered 100 metres, 100 arrived, quality accepted 96 — the bill clears for 96 and not a metre more. The system flags GRN ≠ PO, invoice ≠ accepted × rate, and any invoice arriving before its GRN.
- The vendor scorecard is computed per transaction from real outcomes — accept rate, on-time delivery, rate against market — and it drives the priority order for the next requisition. Sourcing follows performance rather than habit.
- Real stock value moves through purchase and the warehouse continuously; cover that nobody tracks is cover nobody can claim on.

Reads ← Inventory & Catalog · Planning/MRP · Manufacturing · **Writes** → Inventory & Catalog · Accounting & GST · Quality & Compliance

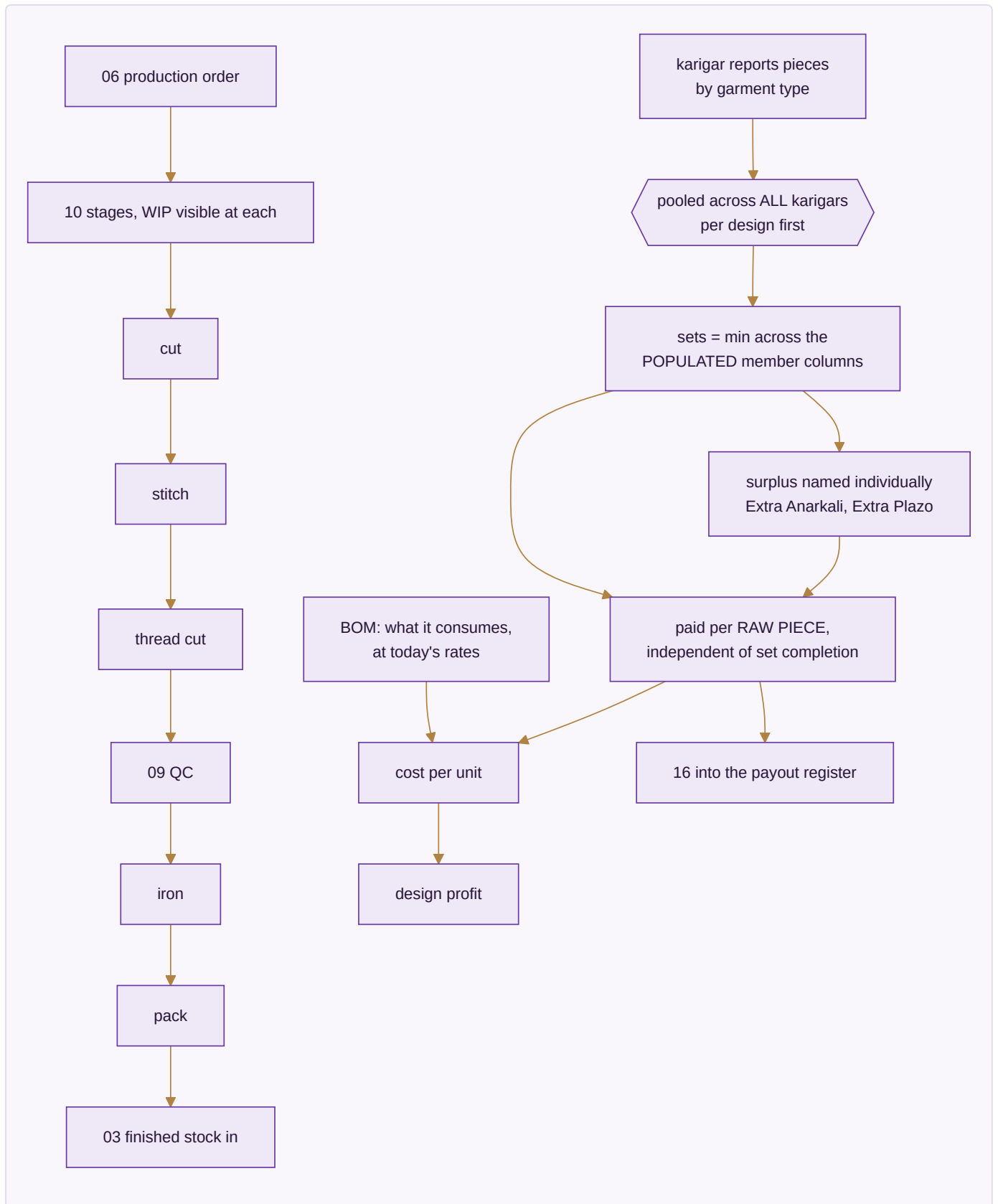
Done when. A vendor invoice for more than was accepted is blocked before payment, and the rejection reason is on the record.

MODULE 08 · MANUFACTURING

Know what a unit really costs to make

What it is. From material in the door to the finished unit — every operation, every worker's earning, and what each design actually cost before it was priced. Design sign-off happens upstream in Module 02; inspection and the compliance record live downstream in Module 09. This module is purely the making.

How it works



The apps (4)

App	State	What it does
Production Orders	SPEC	Your own stages from first operation to finished goods, with work-in-progress visible at each
Piece-rate & Contractors	SPEC	Output-based pay: pooled set completion, per-unit rates, rework and advances resolved into one payout
BOM & Consumption	SPEC	What each design consumes — fabric, zari, lining, trims, packing — costed at today's rates
Maintenance	SPEC	Machines and tools: what is due for service, what it cost, what stopped while it was down

What it owns. `production_orders` · `production_stages` · `bom` · `bom_items` · `karigar_assignments` · `karigar_reports` · `performance_flags`

The karigar costing rules — the heart of this module

These are settled and already proven in a working engine. They are unusual enough to be worth stating exactly, because getting any one of them wrong changes what a person gets paid.

1. **23 garment columns collapse to 13 set types.** A set is a named combination, not an arbitrary grouping.
2. **Pool across all karigars per design first, then apply the rule.** Pooling after applying the rule produces a different, wrong answer.
3. ****Sets = the minimum across the *populated* member columns of that set type.**** A column nobody worked on is not a zero that drags the set count to nothing — it is simply not part of the calculation.
4. **Extras are named individually** — Extra Anarkali, Extra Plazo — never a generic leftover bucket, and there is **no combined "Set + Extra" total column**, because that column would double-count.
5. **Cost is per raw piece, independent of set completion.** A surplus piece that completes no set is still work somebody did, and is still paid.
6. **A missing rate posts ₹0 and raises a flag — never a guessed rate.** A guessed rate is a silent error in someone's wages.
7. Alteration earning is alteration hours × ₹100; an alteration caused by the worker's own mistake is ₹0.
8. A performance flag fires when the same person, design and task takes more than 1.2× the previous hours, and asks why by WhatsApp.

The acceptance gate — figures that must reproduce to the rupee

Designs	Karigar units	Sets	Pieces	Total	Flagged
143	29	25,307	59,110	₹26,90,062	5 designs with no rate

A mismatch against these is a bug, not a rounding difference. The existing Python engine is kept as the independent checker: the ported engine must agree with it to the paise.

Reads ← Purchase · Planning/MRP · Design & Sampling · **Writes** → Inventory & Catalog · HR & Payroll · Accounting & GST · Quality & Compliance

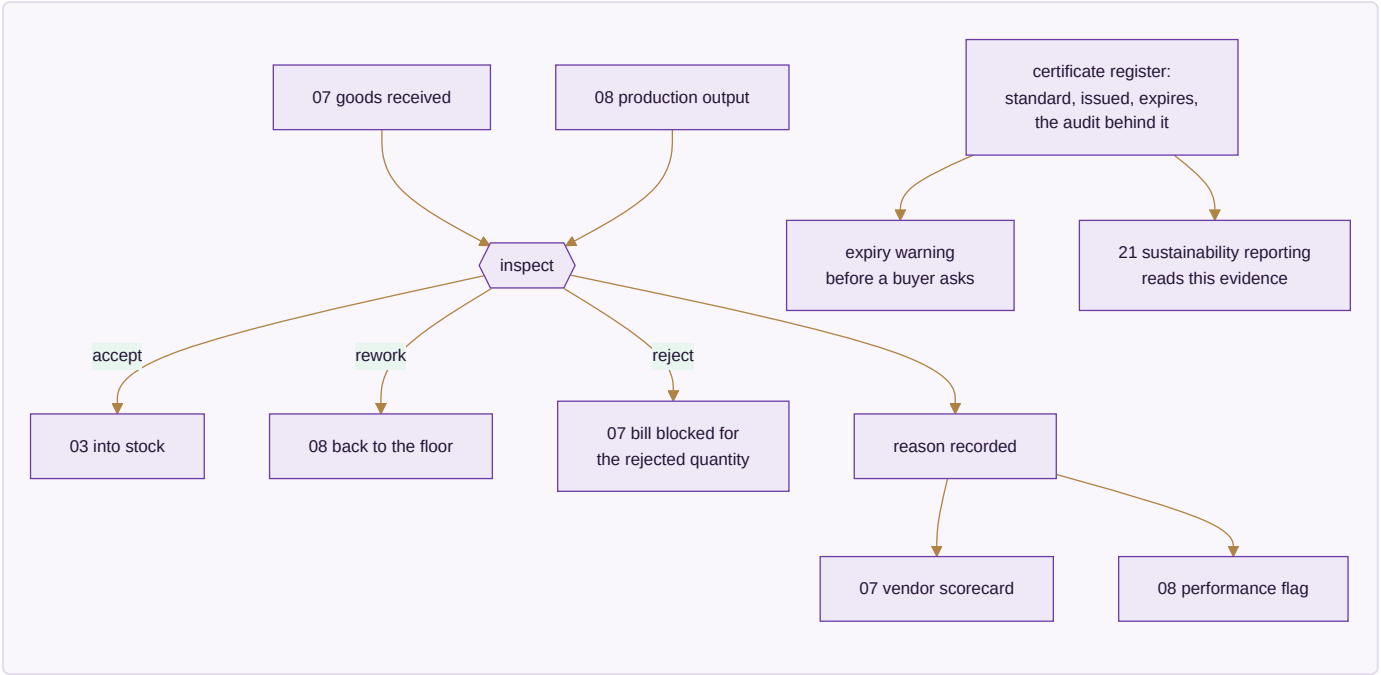
Done when. Three production orders run to completion — self-made, full job work, partial job work — and the acceptance-gate totals reproduce exactly.

MODULE 09 · QUALITY & COMPLIANCE

Certify what was received and what was made

What it is. Inspection used to be one step buried inside Manufacturing. It stands on its own here because rejecting goods at receipt and rejecting goods on the floor are the same discipline — and because the certificates that prove a standard is followed belong beside the inspections that back them up, not scattered across email.

How it works



The apps (2)

App	State	What it does
Quality Control	SPEC	Accept, reject or rework — on goods received and goods made — with reasons that feed the vendor scorecard and the performance flags alike
Certificate & Compliance Register	SPEC	Every standard held, with issue date, expiry and the audit that backs it

What it owns. qc_records · qc_defect_categories · compliance_certificates · compliance_audits

The rules that matter

- A rejection is never just a number — it carries a reason, and that reason is what makes the vendor scorecard and the performance flag meaningful rather than decorative.
- The quantity accepted at QC, not the quantity delivered, is what the three-way match pays against. This is the join between this module and the money.
- A certificate about to lapse is visible *before* a buyer asks for it and finds it expired.
- What this register holds is precisely what a sustainability report is later built from — which is why the reporting app in Module 21 reads it rather than collecting evidence separately.

Reads ← Purchase · Manufacturing · **Writes** → Purchase · Manufacturing · Inventory & Catalog

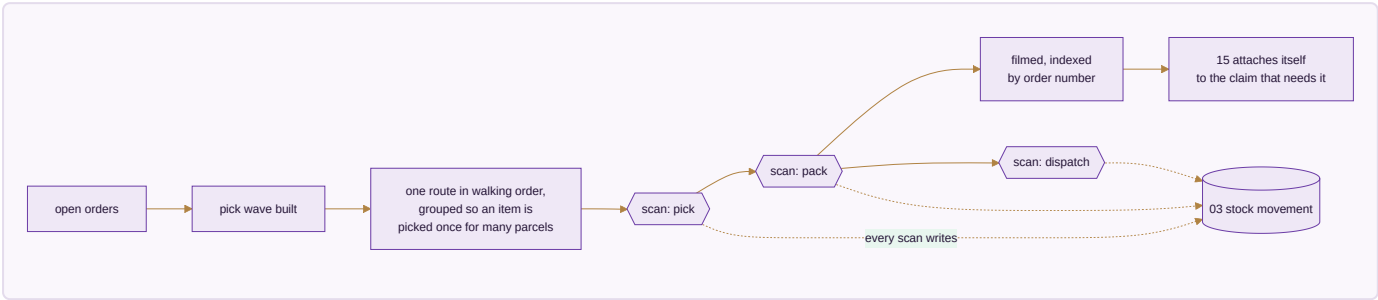
Done when. A rejected receipt blocks payment for exactly the rejected quantity, and an expiring certificate raises a warning before its expiry date.

MODULE 10 · WAREHOUSE

Pick right the first time — and prove what you sent

What it is. Bin-level instructions and barcode scanning so the right item leaves the building and stock stays honest — and a recording of each parcel as it is packed, so an argument about what was in it is settled by footage instead of memory.

How it works



The apps (3)

App	State	What it does
Picking & Bins	SPEC	Pick lists that name the bin, in walking order, so nobody crosses the floor twice
Barcode Operations	SPEC	Scan to pick, pack, dispatch and count from a phone — the same scan whatever channel the order came from
Packing Video	SPEC	Every parcel filmed as it is packed and indexed by order number

What it owns. bins · pick_lists · pick_list_lines · barcode_scans · packing_videos

The rules that matter

- A picker is never sent to an empty rack — the wave is built from the stock record, not from hope.
- **Every scan writes a stock movement.** That is what keeps the running balance real rather than a number someone updates at the end of the day.
- The day is grouped by product, so one design is picked once for eleven parcels instead of eleven times.
- The footage attaches to the claim by order number automatically. A wrong-item claim answered with the clip is a claim won on evidence rather than argument.

Reads ← Sales · E-commerce/OMS · Inventory & Catalog · **Writes** → Inventory & Catalog · Sales · E-commerce/OMS

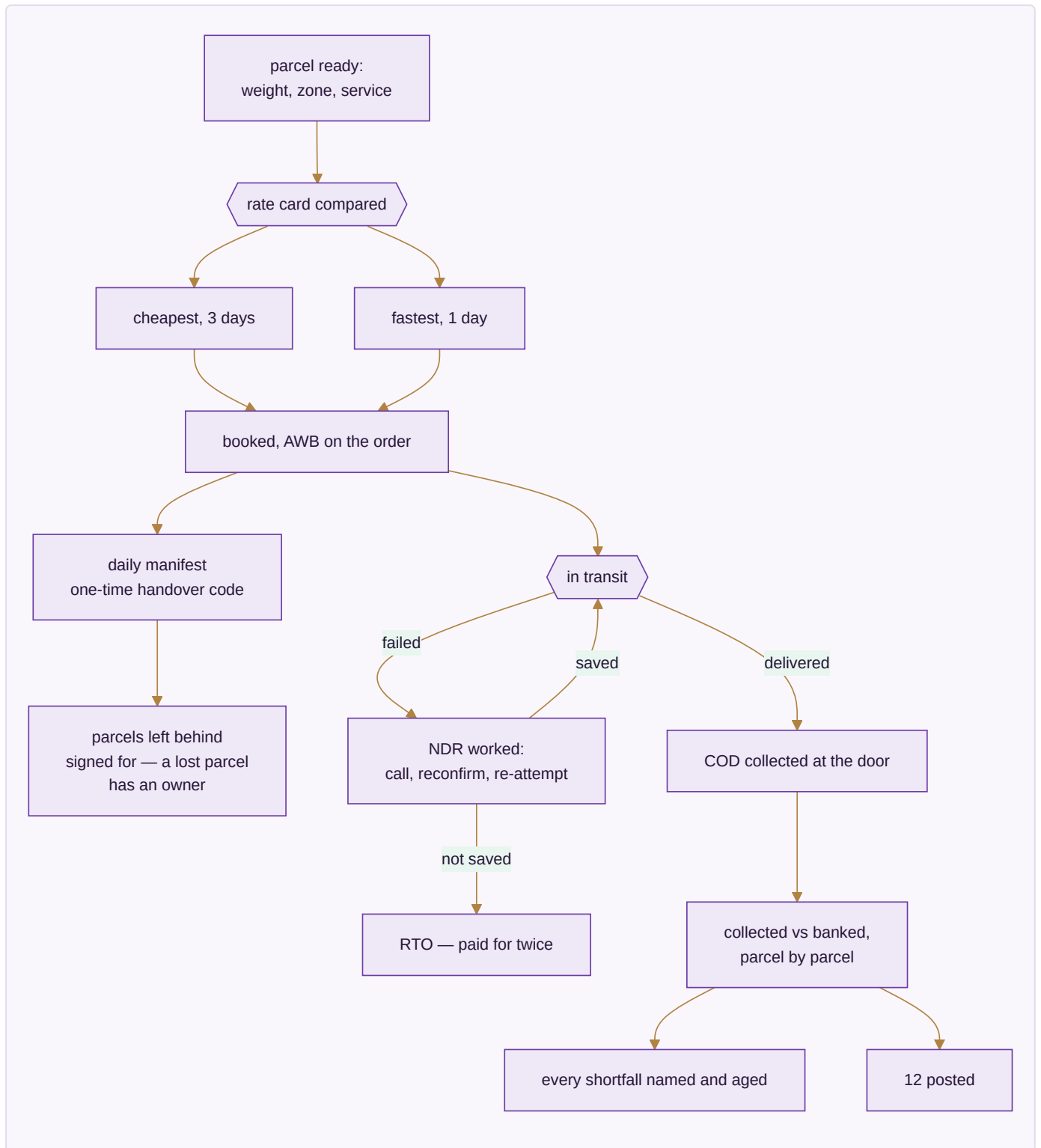
Done when. A parcel is picked from the right bin, filmed, and its clip is already attached when the claim arrives.

MODULE 11 · LOGISTICS

The courier network — rates, failed deliveries and the COD money

What it is. Booking one parcel happens on the order, in Sales. This module is the network behind it: what each courier charges before you pick one, what happens to a delivery that fails, and whether the cash collected at the door actually reached your bank.

How it works



The apps (5)

App	State	What it does
Rates & Zones	SPEC	Every courier's rate card by zone, weight slab and service — cheapest and fastest both known before booking
NDR & RTO Rescue	SPEC	A failed delivery worked while it can still be saved, before it becomes a return you pay for twice
COD Remittance	SPEC	What was collected at the door against what reached the bank, parcel by parcel
Handover & Manifest	SPEC	What went out against what the courier actually took, with a one-time code and a signed record of what was left
Fleet	SPEC	Own vehicles, if any — trips, cost, service due. Optional; most run couriers only

What it owns. `courier_rates` · `shipments` · `ndr_cases` · `cod_remittance` · `manifests` · `vehicles`

The rules that matter

- Both the cheapest and the fastest option are known before booking. Choosing without seeing both is choosing blind.
- **An NDR is worked while it can still be saved.** An RTO costs freight in both directions and returns handled goods — the rescue attempt is almost always cheaper than the return.
- COD is reconciled parcel by parcel, and every shortfall is named and aged. A lump-sum remittance that roughly matches is not reconciliation.
- The manifest carries a signed record of what was *not* taken, so a parcel lost between the packing table and the van has an owner.

Reads ← Sales · E-commerce/OMS · Warehouse · **Writes** → Accounting & GST · Sales · E-commerce/OMS

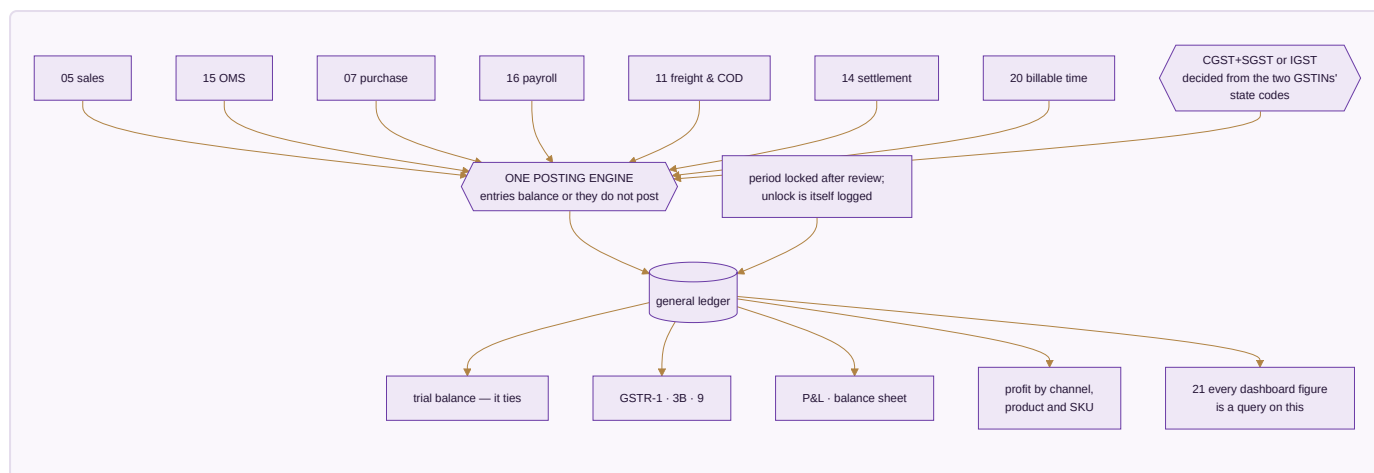
Done when. COD collected reconciles to COD banked parcel by parcel, and an NDR is worked before it turns into an RTO.

MODULE 12 · ACCOUNTING & GST

Books that always balance

What it is. A full double-entry ledger built for Indian compliance — not a tax report bolted onto a spreadsheet. This system keeps the books on its own; no other accounting package is required, ever. Existing packages stay available as connectors for anyone already running one, but no figure in the business is ever *sourced* from them.

How it works



The apps (9)

App	State	What it does
Accounting	SPEC	Chart of accounts, nine voucher types, and the one posting engine every voucher writes through
Invoicing	SPEC	GST tax invoices computed from the lines to the paise, with round-off and e-invoice IRN
Expenses	SPEC	Spend by category with approvals, and bill OCR to save typing
GST & Tax	SPEC	CGST, SGST, IGST, TDS, TCS, input credit and the GSTR returns, filed per registration
ITC Reconciliation	SPEC	Purchases matched against the portal's own GSTR-2A/2B before a return is filed
Receivables, Payables & PDC	SPEC	Payments allocated against named invoices, and post-dated cheques posting on realisation
Fixed Assets & Depreciation	SPEC	The asset register with both straight-line and written-down-value tracked side by side
Year-End Close & Period Lock	SPEC	Carry-forward at year end, and a locked period no backdated edit can touch
Finance Reports	SPEC	P&L, balance sheet, and profit by channel, product and SKU

What it owns. `chart_of_accounts` · `voucher_series` · `journal_entries` · `journal_lines` · `gst_returns` · `gst_input_credit` · `tds_entries` · `tcs_entries` · `bank_accounts` · `bank_transactions` · `fixed_assets` · `depreciation_entries` · `post_dated_cheques` · `bill_allocations` · `period_locks`

The rules that matter

- **One posting engine.** Every voucher writes the ledger through it, and an entry that does not balance does not post — it is refused, not saved half-done.
- **The tax type is derived, never chosen.** Intra-state versus inter-state comes from comparing the two GSTINS' state codes. A hand-picked tax type is a hand-made error.
- Rates default from the HSN and are **versioned by effective date**, so an invoice from last year keeps last year's rate after a rate change.

- Input credit is reconciled against the government's own data before GSTR-3B. Without it, a filing is a guess about how much credit you may legitimately claim.
- **Round-off posts to its own ledger account** — never absorbed into the sale amount, because absorbing it corrupts the GST calculation underneath.
- A closed period locks. No backdated edit without an admin unlock, and the unlock is itself logged.

Build order inside this module — fixed by the nature of the thing: posting engine and audit first, wrapping every write from day one; then sales and purchase invoices, verified against the trial balance; then the other seven voucher types; then year-end close and period lock; then GST returns and 2A/2B reconciliation.

Reads ← every module · **Writes** → Finance Reports · Treasury

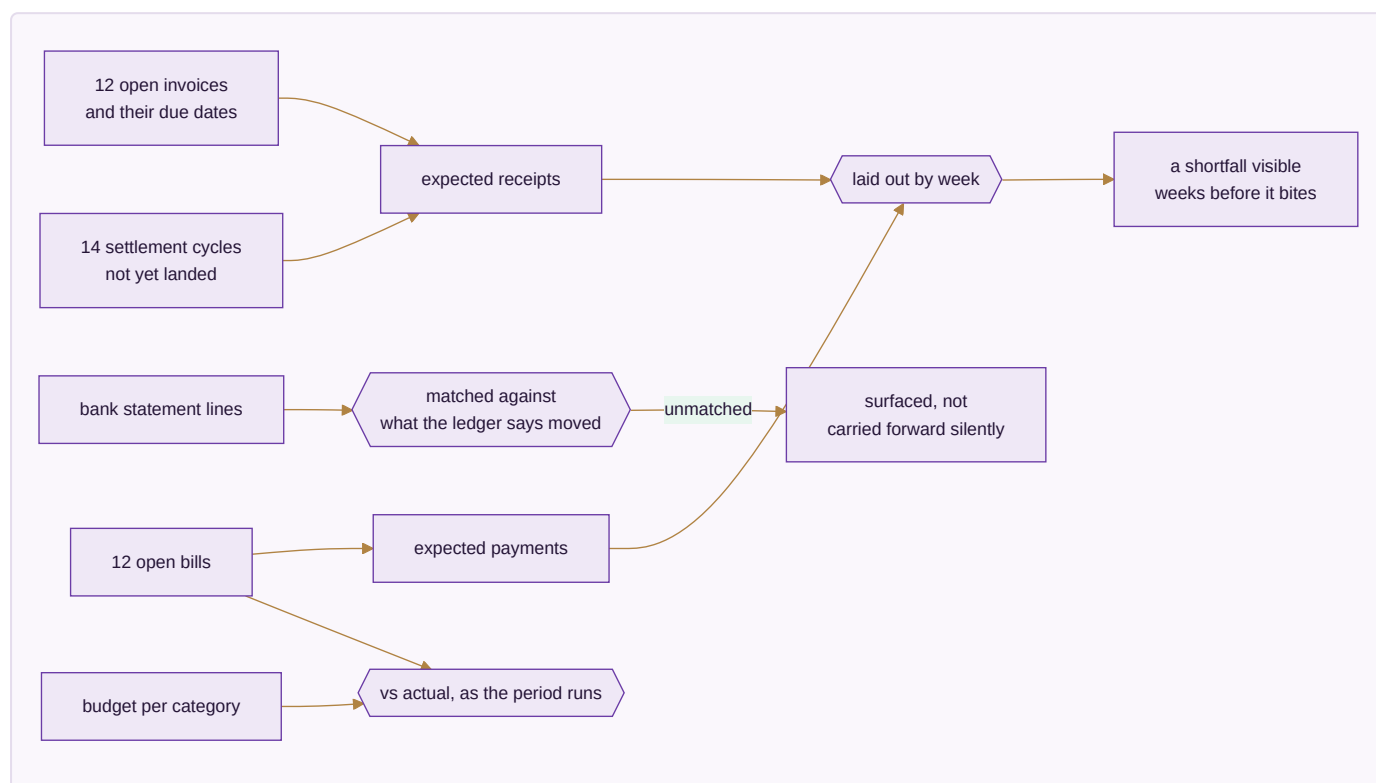
Done when. One month of books closes cleanly, the trial balance ties, and GSTR-1 and GSTR-3B generate and verify.

MODULE 13 · TREASURY & FINANCIAL PLANNING

Know what cash is coming, not just what already arrived

What it is. Accounting records what happened. This module is about what happens next — how much cash is genuinely expected and when, and whether spending against a budget is on track while there is still time to act.

How it works



The apps (3)

App	State	What it does
Cash Flow Forecast	SPEC	Expected receipts and payments by week, drawn from open invoices and bills rather than typed in
Banking & Reconciliation	SPEC	Statement lines matched against the ledger's own record, with anything unmatched surfaced
Budget vs Actual	SPEC	A budget per category and period, tracked against real spend while the period is still running

What it owns. `cash_forecasts` · `bank_reconciliation_sessions` · `budgets` · `budget_lines`

The rules that matter

- The forecast is **derived from open invoices and bills**, not typed in. A forecast someone maintains by hand is out of date the day after it is written.
- Marketplace payouts that have not landed yet are part of expected cash — that is what makes the forecast honest for a business selling on seven channels.
- An unmatched bank line is surfaced, never quietly carried forward. Carried-forward differences are how a reconciliation becomes fiction.
- Budget variance is shown **while the period runs**. A variance discovered after close is history; the only thing left to do with it is explain it.

Reads ← Accounting & GST · Sales · Purchase · **Writes** → Accounting & GST

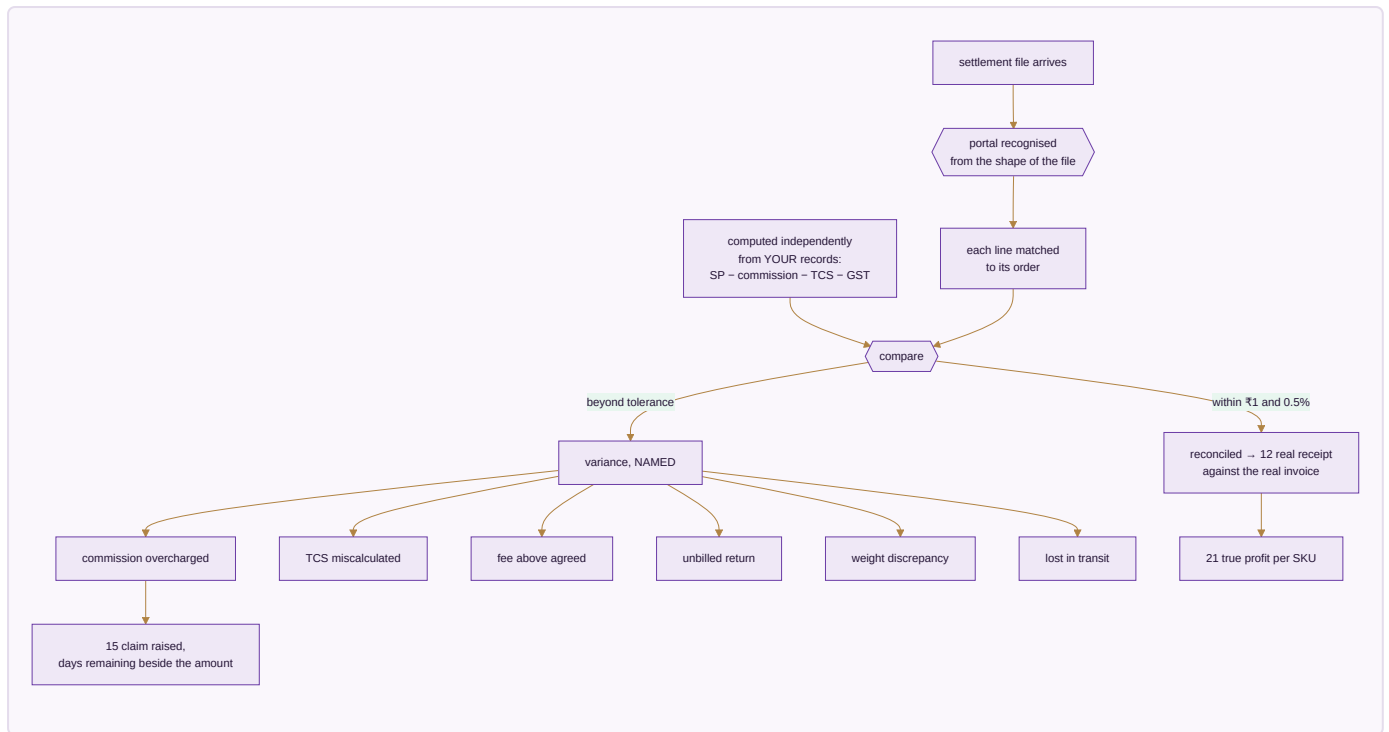
Done when. A twelve-week cash forecast is produced with no manual entry, and a bank statement reconciles with every unmatched line named.

MODULE 14 · SETTLEMENT

Get paid what you are owed — cycle by cycle

What it is. A marketplace does not pay you what the customer paid. It pays selling price minus a commission, minus a collection fee, minus a shipping charge it decided, minus a return it may or may not have handled, minus tax it withheld — and hands you a file with no total you can trust. This module reads that file, works out what each line *should* have been, and puts the two side by side.

How it works



The apps (3)

App	State	What it does
Payout Cycles	SPEC	What each cycle should pay, what actually landed, and on which day — so a late payout is visible the day it is late
Fee & Commission Audit	SPEC	The published rate against the rate actually charged, by category, SKU and tier
TCS & TDS Register	SPEC	Every rupee withheld, matched against the portal's own figures

What it owns. `settlement_cycles` · `fee_audit_lines` · `tcs_tds_register` , reading `marketplace_settlements` from Module 15

The rules that matter

- **The expected figure is computed from your own records, not read back from the file.** The whole point is comparing the file against an answer the file cannot influence.
- **Commission is read from the file, then challenged against the published rate.** A system that assumes the commission is whatever the file says can never detect an overcharge — it has already agreed to it.
- A line is flagged only past **₹1 or 0.5%**, so the variance list holds real problems rather than a thousand one-paise arguments.
- **Every variance has a named kind.** You dispute a weight discrepancy differently from a lost parcel; the name is what makes the claim actionable.
- Days remaining sit beside the rupees at stake. A valid claim that lapses because nobody saw the clock is money given away.

The gate. A real settlement file must reconcile at **98% or better** automatically, and per-SKU profit must land within **₹10** of your own records. Below that the module is not trustworthy enough to run money on — and a settlement tool you cannot trust is worse than none, because it lends confidence to a wrong number.

Reads ← E-commerce/OMS · Accounting & GST · **Writes** → Accounting & GST

Done when. A real settlement file reconciles at 98% or better and every variance it raises is one you agree is genuinely real.

MODULE 15 • E-COMMERCE / OMS

Every marketplace and your own website, one queue

What it is. Stop logging into seven seller panels and your own store admin. Every order lands in one pipeline and one stock number goes back out to all of them — then the money side closes in the same module: what each channel paid, what it kept, what came back, and what it still owes.

How it works



CHANNELS - pulled every 15 min, idempotent by external id

Amazon

Flipkart

Myntra

Meesho

Ajio

Nykaa

JioMart

own storefront

returns triaged

which kind?

customer

₹20 cost

courier

₹5 cost

wrong

full selling price,
DEAD STOCK, never restocked

sorted by TIME REMAINING,
not time received

a 12h Amazon order placed at
2pm
outranks a 48h Ajio order from
noon

allocation desk:
which warehouse can actually
serve it

none can

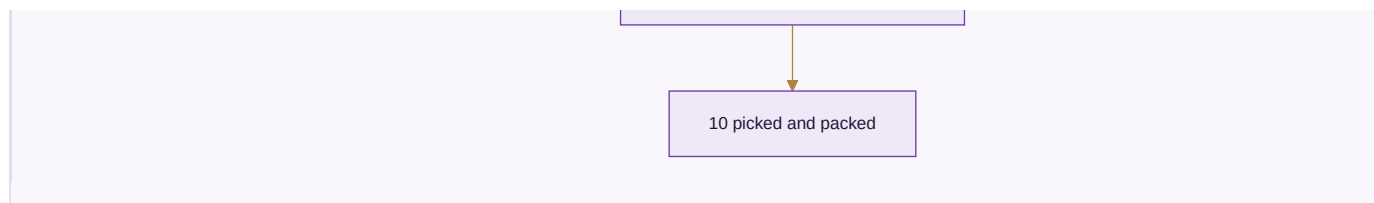
no date promised —
needs a purchase or production
order

can

03 stock reserved

labels printed in a batch,
never uploaded outside to be
cropped

repeat pattern flagged



The apps (11)

App	State	What it does
Marketplace OMS	BUILT	Every channel in one order queue, with the right cut-off counting down on each order
Order Management	BUILT	One pipeline new to delivered, with the allocation desk and the promise-date engine
Manual Data Check	SPEC	Upload the sheets you already download and read ten cross-checks back, every figure clickable to the transactions behind it
Reconciliation	SPEC	Match every payout to the order line that earned it, and expose the gap
Claims & Disputes	SPEC	Shortfalls, weight disputes and lost parcels filed with evidence, answered before the clock runs out
Returns / RMA	SPEC	Customer, courier and wrong returns kept apart — only one of the three is really your fault
Channels & Storefronts	SPEC	Connect a channel once and it stays in step: catalogue out, price out, stock out, orders in
Labels & Documents	SPEC	The channel's PDF turned into something a packing table can work from, in one batch
Listing & Catalog Manager	SPEC	Bulk create and edit listings across channels, catching listed-but-out-of-stock mismatches
Size / Fit Recommendation AI	SPEC	A fit suggestion from the item's measurements and the return history by size
AR / Virtual Try-On	SPEC	Seeing drape and colour before buying, where a flat photo leaves too much to guess

What it owns. `marketplace_orders_raw` · `marketplace_settlements` · `marketplace_settlement_lines` · `returns` · `claims` · `channels` · `channel_listings`

The rules that matter

- **The queue sorts by time remaining, not time received.** Channels have different dispatch windows; sorting by arrival breaches the tight ones while the loose ones sit safe.
- Orders are pulled on a schedule and are **idempotent by external ID**, so the same order never doubles no matter how often the pull runs.
- **An order no warehouse can serve gets no promised date.** It needs a purchase or a production order, and nothing is gained by giving the customer a date in the meantime.
- **A wrong return is dead stock and is never restocked.** Booking it back into sellable stock is how a business sells the same damaged piece twice and loses the customer permanently.

- **Labels are never uploaded to an outside website to be cropped.** That is a customer's name and address leaving your control for a convenience.
- Commission is read from the settlement file, never assumed — the challenge to it happens in Module 14.

Reads ← Inventory & Catalog · CRM · Sales · Accounting · Logistics · Settlement · **Writes** → Inventory & Catalog · Accounting · Warehouse · Logistics · Settlement

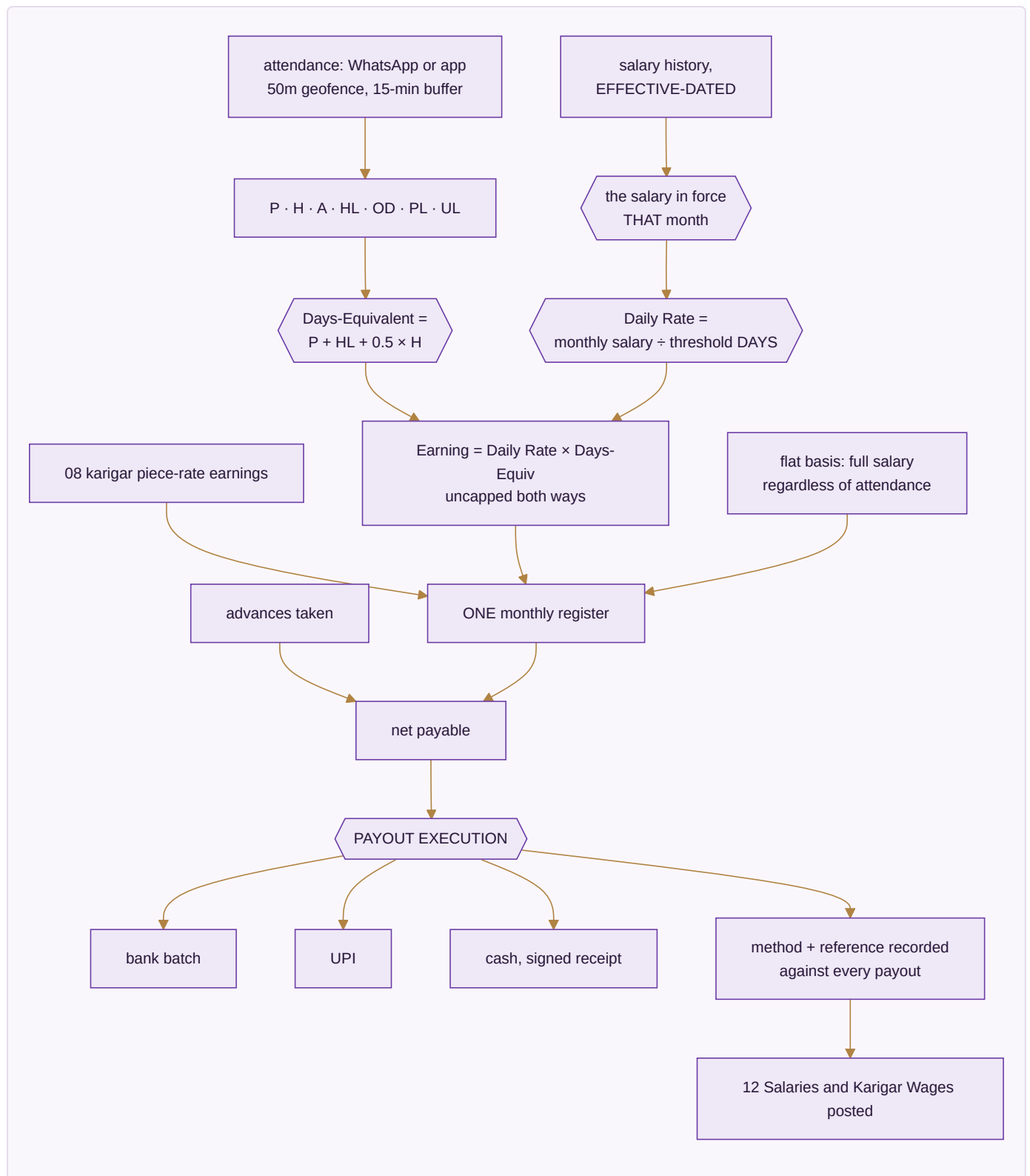
Done when. A full week runs with every channel live, settlements reconciled, and no panel oversells.

MODULE 16 · HR & PAYROLL

Pay people right, on time

What it is. Salaries and output-based earnings in one register, with attendance driving both — whether people are on a monthly wage, an hourly rate, or paid by what they finish. And, unlike the version of this that stops at a calculated figure, it carries through to the money actually leaving the business.

How it works



The apps (4)

App	State	What it does
Staff & Contractors	SPEC	Attendance, effective-dated salary and output-based earnings in a single register
Time-off & Advances	SPEC	Leave, festival advances, and exactly how they change this month's payout
Appraisal & Hiring	SPEC	Performance reviews and a hiring pipeline that ends in an employee record
Payout Execution	SPEC	Where the calculation becomes money leaving the business, with method and reference on every payout

What it owns. `staff_salary_history` · `attendance` · `eod_reports` · `leave_requests` · `advance_requests` · `payroll_runs` · `payroll_slips` · `karigar_earnings_summary` · `piece_rates` · `task_threshold_rates` · `payout_batches`

The pay rules, exactly

1. Attendance codes are **P · H · A · HL · OD · PL · UL**.
2. **Days-Equivalent = P + HL + 0.5 × H**. A holiday pays a full day.
3. **Daily Rate = the resolved monthly salary ÷ the resolved threshold DAYS** — both effective-dated.
4. **Earning = Daily Rate × Days-Equivalent, uncapped in both directions**. Working beyond the threshold earns beyond the salary; working under it earns under.
5. A flat-basis worker draws full salary regardless of attendance. A piece-rate worker is hours × flat rate with no attendance row at all.
6. Three month-states are distinguished and never conflated: **Not employed**, **No data**, and a real month. Treating the first two as zero is how someone gets paid nothing for a month they worked.

Worked example. Salary ₹15,000 with a 26-day threshold gives a daily rate of ₹576.92. A month of 24 P, 1 HL and 2 H is $24 + 1 + 1.0 = 26.0$ days-equivalent, so ₹15,000.00 exactly. The same person on 28.0 days-equivalent earns ₹16,153.85 — it scales past the threshold because the rule is uncapped.

The rule that closes the loop. A salary edit is made **once, with an effective-from date**. The previous row closes automatically, past months keep their old rate, and a future-dated raise activates itself when that month's payroll runs. No historical payroll is ever rewritten.

Reads ← Manufacturing · **Writes** → Accounting & GST

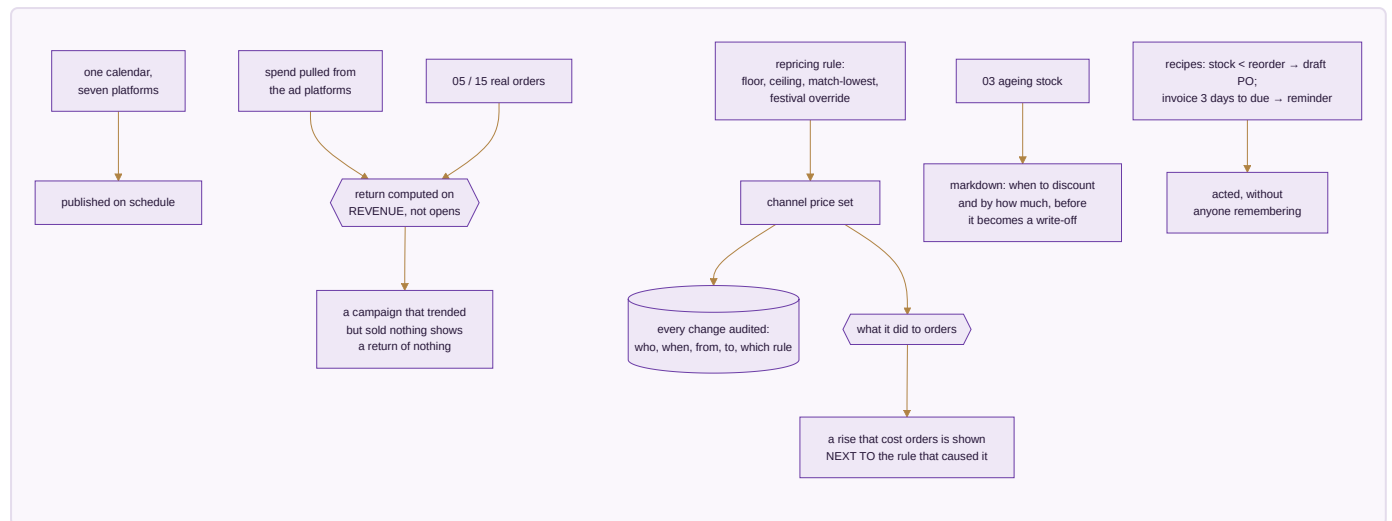
Done when. A full month's payroll runs end to end with zero manual touch, reconciles to the owner's own figures, and every rupee paid has a method and a reference against it.

MODULE 17 · MARKETING

Sell more without discounting

What it is. The demand-generation side of the same order book. Its discipline is that every lever it pulls is judged by what it did to orders and margin — using the same stock number and the same ledger as everything else.

How it works



The apps (7)

App	State	What it does
Social Calendar	SPEC	Plan and publish across every platform from one calendar
Campaigns	SPEC	Campaigns measured on real revenue, not opens
Repricing Engine	SPEC	Rules per channel and SKU, every change audited, and what each one actually did
Automation	SPEC	If this happens, do that — across any module, without writing code
Blog & Pages	SPEC	Articles and landing pages published to your own site with meta and internal links set
Events	SPEC	Trade shows worked as a channel, leads landing straight in CRM
Markdown / Clearance Optimization	SPEC	The repricing engine aimed at ageing stock before it becomes a write-off

What it owns. `content_calendar` · `campaigns` · `influencers` · `asset_library` · `repricing_rules` · `automation_recipes` · `events` · `markdown_plans`

The rules that matter

- **Return is computed on revenue.** Opens, clicks and reach describe attention, not money.
- **A price that rose and took orders down with it is shown as exactly that, beside the rule that raised it.** Pricing is the one lever whose damage is invisible unless you look for it — you simply sell fewer, without being told why.
- Every price change is audited: who, when, from what to what, by which rule. A price is money; an unexplained change to it is the same failure as an unexplained ledger entry.

- **Repricing sets price and only price.** It cannot touch quantity, so no pricing automation can ever oversell by moving a number it had no business touching.
- Discounting ageing stock is a decision with a right time. Too early gives away margin that was available; too late turns a sale into a write-off.

Reads ← Inventory & Catalog · CRM · **Writes** → Sales · E-commerce/OMS

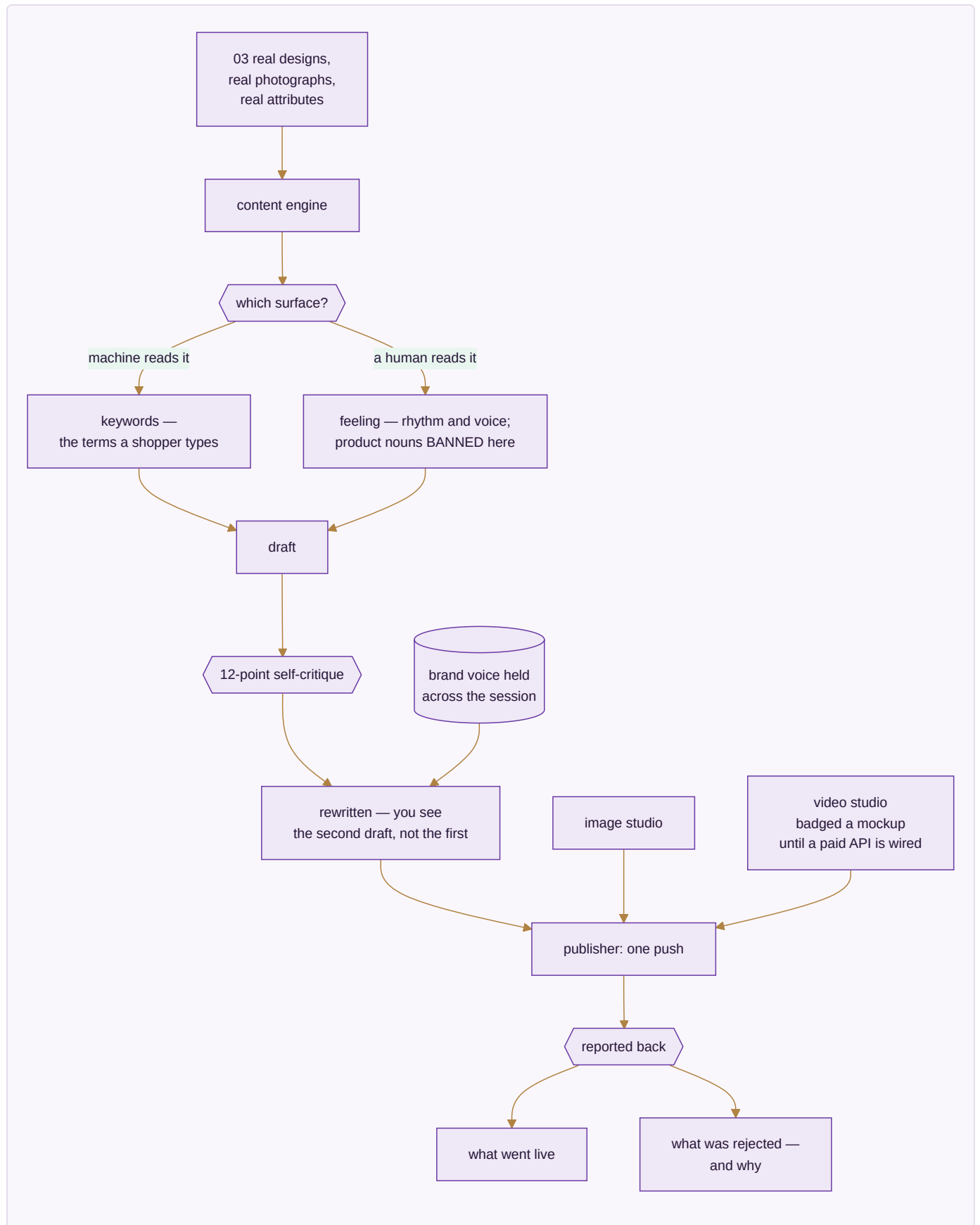
Done when. A month of content publishes on schedule, and a repricing rule can be judged by what it actually did to orders.

MODULE 18 · AI CONTENT ENGINE

Write it, shoot it, cut it — from the catalogue you already have

What it is. The module the platform is named for, deliberately at eighteen rather than first: it produces content *about* the catalogue, so the catalogue has to exist and be correct before there is anything true to write about. Being wired to the same database as the shop floor is what stops it inventing a product that is not real.

How it works



The apps (5)

App	State	What it does
Content Engine	SPEC	Fourteen stages in your own voice, each written from your own catalogue so the words match the thing
Image Studio	SPEC	Layers, free transform, background removal, channel presets and alt text — a phone photo becomes a channel-compliant product image
Video Studio	SPEC	Text and image to video, reels and ad cuts sized per channel
Design Studio	SPEC	A full design surface exporting at whatever size the channel or printer asks for
Publisher	SPEC	One push everywhere, reporting what went live and what was rejected, with the reason

What it owns. `ai_runs` · `ai_listings` · `ai_design_analytics` · `asset_projects`

The rules that matter

- **Structured data gets keywords; anything a human reads gets feelings.** A back-end field is written for a search algorithm; a caption is written for a person. Writing both the same way is what makes machine-written content read as machine-written.
- **Product nouns are banned from creative surfaces.** The right phrase in a search field is exactly the wrong phrase in a caption, and that bleed is the clearest tell of lazy automation.
- **The engine criticises its own draft before showing it.** A model's first attempt is rarely its best; building the criticism into the pipeline is what clears the bar of publishable rather than demonstrable.
- **Generation stays badged a mockup until a real paid API is wired.** Showing a simulated render as a finished one is exactly the dishonesty this whole platform is built to avoid, so the label is not optional.
- A publish that silently fails on two of six channels leaves you believing you are present where you are not. The report closes that gap.

Reads ← Inventory & Catalog · **Writes** → Marketing · E-commerce/OMS

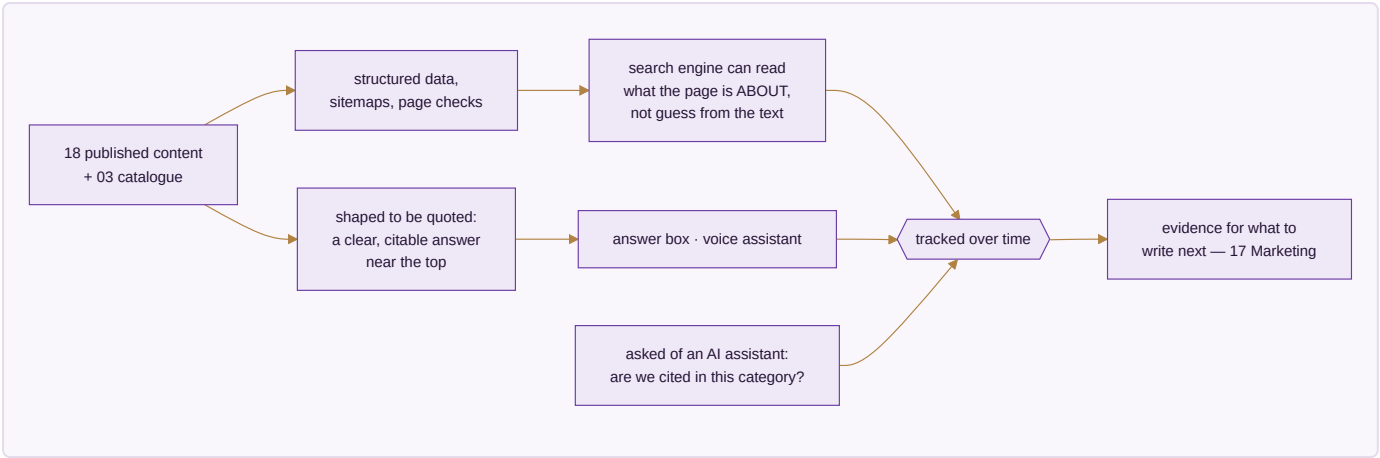
Done when. A listing generates for one design across six platforms in under twenty seconds and publishes, with every rejection explained.

MODULE 19 · SEO, AEO & AIO

Be found by a search box, an answer box, and an AI

What it is. Content exists by the time this module is reached; here it is made findable — by a traditional search engine, by the answer box above the results, and by the AI assistants now answering shopping questions directly instead of sending someone to a results page.

How it works



The apps (3)

App	State	What it does
Technical SEO & Schema	SPEC	Structured data, sitemaps and page-level checks so a search engine can read what a page is about
Answer-Engine Optimization	SPEC	Content shaped to be quoted directly rather than written only to be scrolled
AI-Engine Visibility Tracking	SPEC	Whether this business is actually cited when someone asks an AI a shopping question in this category

What it owns. `seo_audits` · `schema_templates` · `answer_blocks` · `visibility_checks`

The rules that matter

- Structured data is how a page stops being guessed at. Prose alone leaves the engine inferring.
- An answer engine quotes; it does not summarise a page written to be scrolled. Content has to be shaped for the quote to exist.
- **AI visibility is rank tracking aimed at a newer kind of results page.** The discipline is the same — measure over time, act on the trend — and ignoring it means being absent from where a growing share of shopping questions now get answered.

Reads ← Inventory & Catalog · AI Content Engine · **Writes** → Marketing

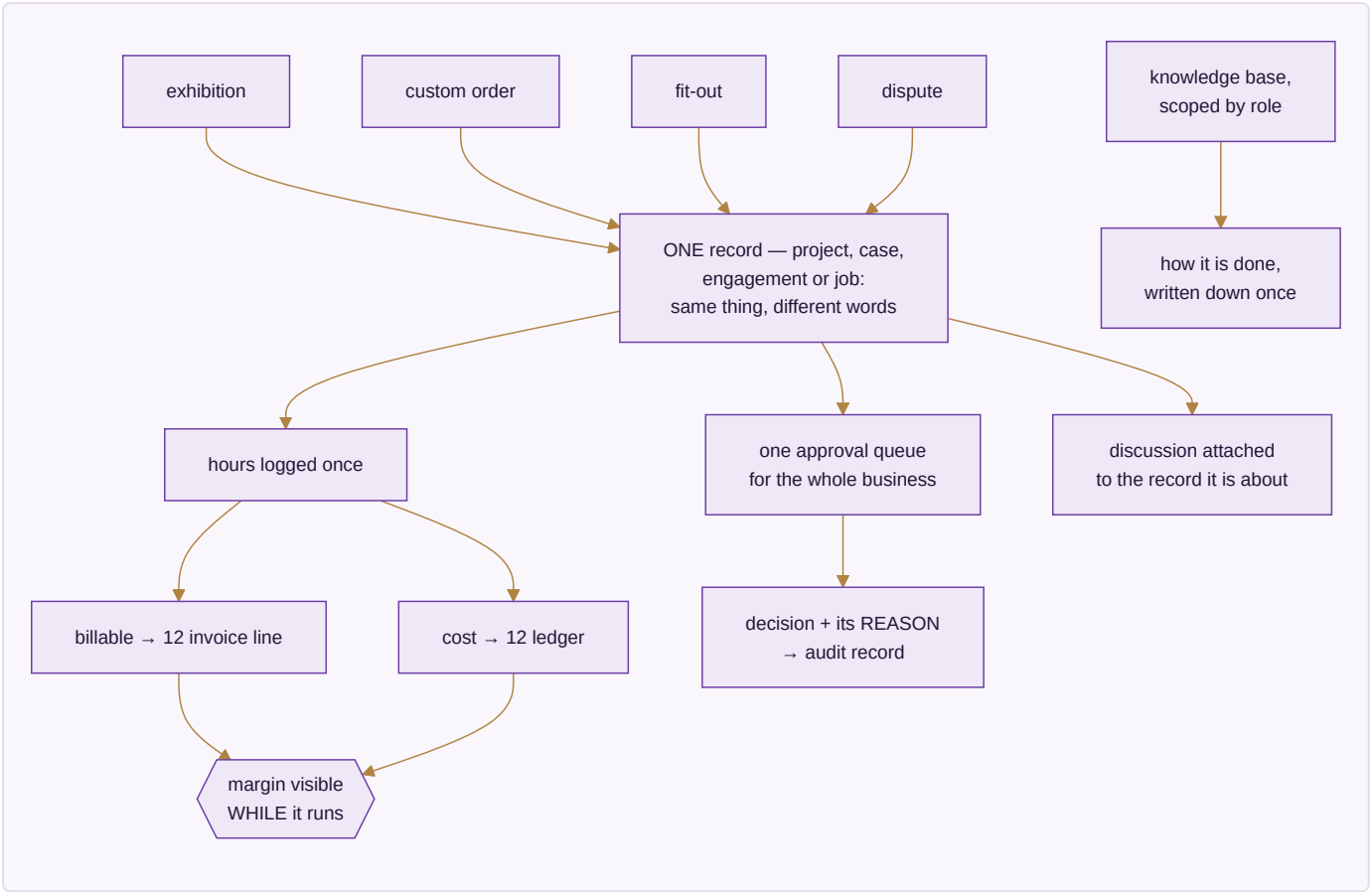
Done when. Every product and content page carries valid structured data, and citation in this category is tracked over time rather than assumed.

MODULE 20 · PROJECTS & COLLABORATION

The work that is not an order — and the talking around it

What it is. Not everything a business does is an order. An exhibition, a custom bulk enquiry that runs six weeks before it becomes an invoice, a godown fit-out, a dispute with a supplier. That work has a deadline, a cost and documents, and it belongs on the same records as everything else.

How it works



The apps (6)

App	State	What it does
Projects & Cases	SPEC	Stages you define, owners, deadlines, documents, billable time and real cost, on one record the ledger can see
Timesheets & Planning	SPEC	Hours against a project or a machine, billable and non-billable kept apart
Approvals	SPEC	One queue for everything waiting on a yes, with the rule that sent it there beside it
Forum	SPEC	Questions and answers that outlive a chat
Discuss	SPEC	Conversation attached to the record it is about
Knowledge Base	SPEC	A role-scoped wiki of standard operating procedures

What it owns. projects · timesheets · approvals · forum_posts · discussions · knowledge_base

The rules that matter

- **Hours are entered once** and become both an invoice line and a cost. Re-keying billable hours from a timesheet into an invoice is exactly where hours get lost and margin quietly leaks.
- **Margin is visible while the work runs.** A margin discovered at the end is a margin you could not steer.
- **Every approval decision carries its reason into the audit record.** A year later, "why did we approve this" has an answer sitting next to the approval, not in someone's memory.
- One approval queue, not one per module — because approvals scattered across modules are approvals that stall unseen.

Reads ← CRM · Sales · HR & Payroll · Inventory & Catalog · **Writes** → Accounting & GST · HR & Payroll · CRM

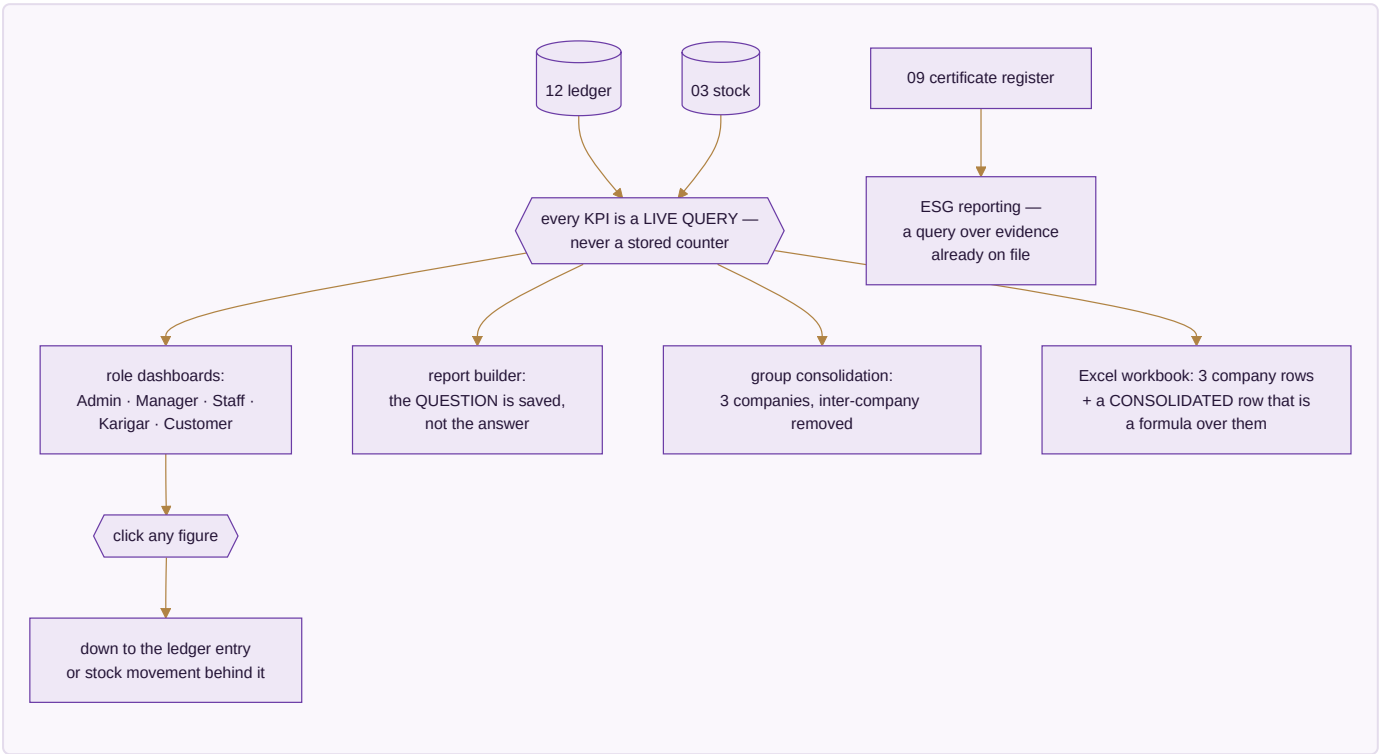
Done when. Billable time on a case becomes an invoice and a cost without being re-keyed once.

MODULE 21 · DASHBOARD & BI

See the whole business without asking anyone

What it is. Every number in the business on one screen, as work happens. It is built last for a reason: it has nothing true to show until the other twenty are producing real records. Two dials govern every screen — which period, and which company.

How it works



The apps (5)

App	State	What it does
CEO Dashboard	BUILT	Cash, sales, stock, profit and alerts on one screen, refreshed as work happens
Report Builder	BUILT	Drag the fields you want; the question is saved, so next month it answers for next month
Group Consolidation	BUILT	Three companies as one set of figures, inter-company entries removed
Excel Dashboard Builder	SPEC	The full workbook, each company a row and the consolidated row a live formula over them
ESG / Sustainability Reporting	SPEC	Water, chemical compliance, waste and packaging, reported from the certificate and audit records already held

What it owns. Nothing. It reads. It stores only saved report definitions and dashboard layouts.

The rules that matter

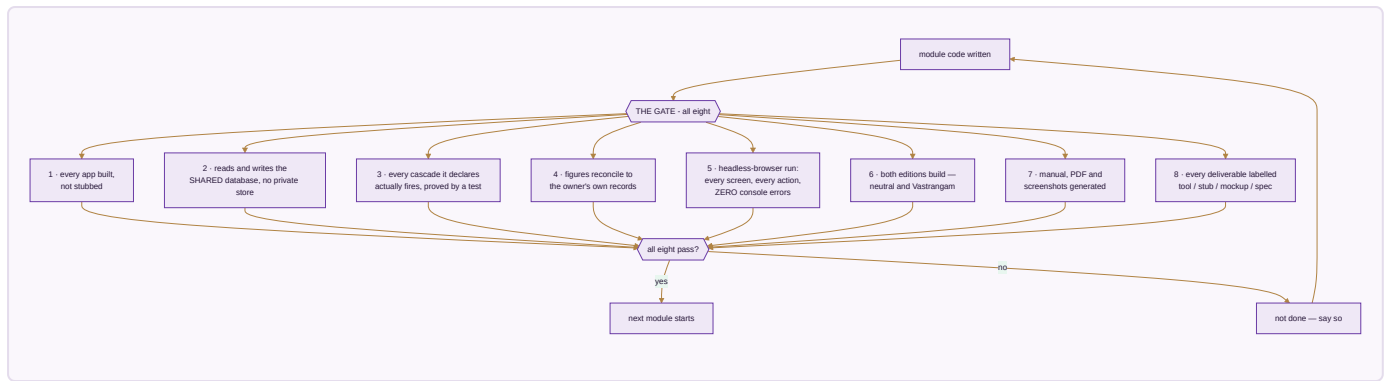
- **Every KPI is a live query over the ledger and stock.** There are no stored counters, because a stored counter is a number that can drift from the truth it claims to summarise.
- **The consolidated row is a formula over the three company rows,** never a separately calculated fourth number — so it foots by construction and an auditor can trust it.
- Group profit removes inter-company sales and purchases. Money the group moved from one pocket to another is not turnover.
- **Any figure clicks down to the record behind it.** A dashboard you cannot interrogate is a dashboard you eventually stop believing.
- The financial year is detected from the data. Nothing is hardcoded.

Reads ← every module · **Writes** → nothing

Done when. Any figure on any screen clicks down to the ledger entry or stock movement behind it, in both editions.

PART III — WHEN IS A MODULE DONE

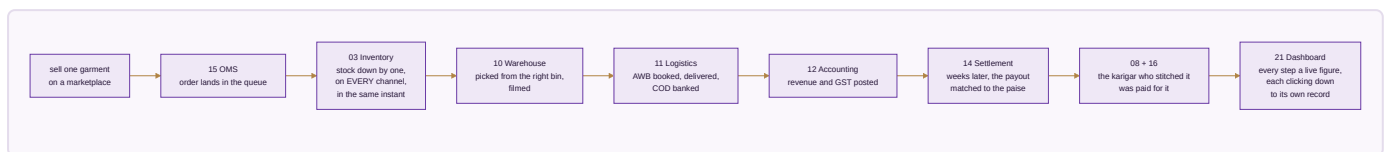
A module is not finished because its code is written. It is finished when all eight of these hold.



Rule 8 is the one that matters most, because it is the one that is easiest to quietly skip. Nothing is called finished that is not. A mockup is labelled a mockup even when a working version would be more impressive to show.

PART IV — THE PROOF

The test of whether this is one system or twenty-one programs sharing a login is a single garment followed end to end. This is re-run at every module boundary.



One transaction. Eight modules. One database.

Everything in this document exists to make that single sentence true. The honest state today is that of the eight modules named in that sentence, sixteen apps are working and the shared database beneath them is written and tested but not yet carrying them — which is the first job of Module 01, and the reason the build order starts where it does.

Every figure in this plan traces to a source: the module and app counts are read directly from the canonical module list the website and every generated document also read, so they cannot drift; the built-versus-spec marks correspond to sixteen apps that pass their own self-tests and a full click-through audit in both editions; and the acceptance figures in Module 08 and the gate in Module 14 are the owner's own numbers, not targets invented here.